

#1 · Steel Structures

What is the default ASTM grade and F_y for wide flange (W-shape) structural steel members on the SE exam?

Answer

A992 is the default grade for W-shapes, with $F_y = 50$ ksi and $F_u = 65$ ksi. If a problem lists a W-shape with no stated grade, assume A992 unless told otherwise.

#2 · Steel Structures

What grade and yield strength is typically used for plates, angles, and channels?

Answer

These shapes are commonly A36 steel, with $F_y = 36$ ksi and $F_u = 58$ ksi. Knowing this by heart saves time versus looking it up in the handbook material table every time.

#3 · Steel Structures

What grades are commonly used for HSS shapes, and what are their yield strengths?

Answer

Round HSS often uses A500 Grade C with $F_y = 46$ ksi, and rectangular HSS often uses A500 Grade C with $F_y = 50$ ksi. A572 Grade 50 is also common for higher strength members.

#4 · Steel Structures

What phi and omega values apply to most yielding and flexure limit states in AISC 360?

Answer

For most yielding and flexure checks, $\phi = 0.90$ for LRFD and $\omega = 1.67$ for ASD. These are the most frequently used factors on this exam so they should be memorized cold.

#5 · Steel Structures

What phi and omega values apply to most fracture and connection limit states?

Answer

For most fracture and connection checks, $\phi = 0.75$ for LRFD and $\omega = 2.00$ for ASD. Mixing these up with the yielding factors is a common source of careless errors under time pressure.

#6 · Steel Structures

What is the fundamental difference between LRFD and ASD as design philosophies?

Answer

LRFD applies a phi factor less than 1.0 that reduces the nominal capacity, while ASD divides the nominal capacity by an omega factor greater than 1.0. Both are set up in AISC 360 Chapter B and the exam can ask a problem using either method.

#7 · Steel Structures

Which AISC chapter governs tension member design?

Answer

Tension members are governed by AISC Chapter D. Two competing limit states must always be checked: gross section yielding and net section fracture.

#8 · Steel Structures

What equation and factors govern gross section yielding of a tension member?

Answer

Gross section yielding uses $P_n = F_y \text{ times } A_g$, with $\phi = 0.90$ for LRFD or $\omega = 1.67$ for ASD. This limit state checks the full cross section away from any connection.

#9 · Steel Structures

What equation and factors govern net section fracture of a tension member?

Answer

Net section fracture uses $P_n = F_u \times A_e$, with $\phi = 0.75$ for LRFD or $\omega = 2.00$ for ASD. A_e is the effective net area, which accounts for both hole removal and shear lag.

#10 · Steel Structures

How is effective net area A_e calculated for a tension member?

Answer

A_e equals $U \times A_n$, where A_n is the net area after subtracting bolt holes and U is the shear lag factor. U comes from Table D3.1 based on how the load is delivered into the member.

#11 · Steel Structures

What does the shear lag factor U actually depend on?

Answer

U depends on the connection configuration, meaning how the load is delivered into the member, not simply the member's shape. A W-shape connected only through its flanges has a different U than one connected through both flange and web.

#12 · Steel Structures

Why do candidates commonly miss the shear lag factor on tension member problems?

Answer

Table D3.1 lists U by connection type, but the exam exhibit does not label which configuration applies. Candidates must recognize the connection geometry themselves before they can look up the right U value.

#13 · Steel Structures

In a bolt hole net area calculation, why is 1/8 inch added to the nominal bolt diameter?

Answer

AISC design practice adds 1/16 inch for damage plus 1/16 inch for standard hole oversize, totaling 1/8 inch, to the nominal bolt diameter to get the hole diameter used in net area calculations. A 7/8 inch bolt therefore uses a 1 inch hole diameter.

#14 · Steel Structures

For a tension member with holes in both flanges at one cross section, how many hole deductions are subtracted from A_g to get A_n ?

Answer

If there are two holes per flange and two flanges, that is four total hole deductions at that cross section. Each deduction equals hole diameter times the local material thickness at that hole.

#15 · Steel Structures

A W10x33 tension member has $A_g = 9.71 \text{ in}^2$, four hole deductions of 1.0 in by 0.435 in flange thickness. What is A_n ?

Answer

$A_n = A_g \text{ minus } 4 \text{ times } (1.0 \text{ times } 0.435) = 9.71 \text{ minus } 1.74 = 7.97 \text{ in}^2$. This net area is then multiplied by the shear lag factor U to get A_e .

#16 · Steel Structures

For that same W10x33 member with $U = 0.90$, what is A_e ?

Answer

$A_e = U \text{ times } A_n = 0.90 \text{ times } 7.97 = 7.17 \text{ in}^2$. This effective net area feeds into the fracture limit state, $P_n = F_u \text{ times } A_e$.

#17 · Steel Structures

For the W10x33 example, which limit state governs, yielding or fracture, and why does that pattern recur on this exam?

Answer

Fracture governs at $\phi P_n = 349.6$ kips versus yielding at $\phi P_n = 437.0$ kips. This happens repeatedly because fracture uses a smaller effective area combined with a lower ϕ factor of 0.75, so it frequently beats yielding even though F_u is larger than F_y .

#18 · Steel Structures

What is the standard approach to checking a tension member's capacity on this exam?

Answer

Always calculate both the gross yielding capacity and the net fracture capacity, then take the lower of the two governing values. Never assume yielding automatically governs just because F_y is smaller than F_u .

#19 · Steel Structures

Which AISC chapter governs compression member (column) design?

Answer

Column design lives in AISC Chapter E. The controlling concept is slenderness, expressed as KL/r .

#20 · Steel Structures

What do the terms in the slenderness ratio KL/r represent?

Answer

K is the effective length factor based on end restraint conditions, L is the unbraced length of the column, and r is the radius of gyration about the axis being checked. A larger KL/r means a more slender, weaker column.

#21 · Steel Structures

Why must both the strong axis and weak axis KL/r be checked for a column?

Answer

The column must be designed for whichever axis has the larger KL/r , since that is the axis that buckles first and governs the lowest capacity. Checking only one axis can miss the true governing failure mode.

#22 · Steel Structures

What K factor applies to an ideal pinned-pinned column?

Answer

$K = 1.0$ for a pinned-pinned column. This is the baseline effective length factor most commonly assumed in braced frame problems.

#23 · Steel Structures

What K factor applies to an ideal fixed-fixed column, and what real-world caveat applies?

Answer

$K = 0.5$ for fixed-fixed in an ideal world, but true full fixity rarely exists in real construction. This value is a theoretical lower bound rather than a typical design assumption.

#24 · Steel Structures

What K factor applies to a fixed-free cantilevered column?

Answer

$K = 2.0$ for a fixed-free column, meaning fixed at the base and free (unrestrained) at the top, such as a cantilevered column. This is the largest common K value and produces the least capacity for a given length.

#25 · Steel Structures

What K factor range applies to a fixed-pinned column?

Answer

$K = 0.7$ to 0.8 for a fixed-pinned column, meaning fixed at one end and pinned at the other. This falls between the pinned-pinned and fixed-fixed extremes.

#26 · Steel Structures

What is the conceptual difference between braced frame and moment frame columns regarding K?

Answer

Braced frame columns have sidesway inhibited, giving K around 1.0 or less. Moment frame columns have sidesway uninhibited, giving K greater than 1.0 , since the frame relies on column bending stiffness to resist lateral movement.

#27 · Steel Structures

Where do K factors for irregular framing conditions typically come from on the real exam versus in practice?

Answer

In practice, K factors come from the alignment charts (nomographs) in the AISC Commentary. On the actual exam, problems almost always give a stated K value or a clearly identified braced versus moment frame condition instead of requiring a nomograph read, since that process is too slow.

#28 · Steel Structures

What splits column behavior into elastic versus inelastic buckling in Chapter E?

Answer

Chapter E compares the column's KL/r against a slenderness limit tied to F_y and E . Above that limit the column is slender and elastic buckling governs; below it the column is stockier and inelastic buckling governs.

#29 · Steel Structures

What is the practical difference between a slender column and a stocky column in terms of capacity?

Answer

A slender column has large KL/r , lower capacity, and its strength is controlled by elastic buckling. A stocky column has small KL/r , and its capacity approaches the full yield capacity F_y times A_g since yielding, not buckling, controls.

#30 · Steel Structures

What is the equation for elastic critical buckling stress F_e used in AISC Chapter E?

Answer

$F_e = \pi^2 E / (KL/r)^2$. This elastic buckling stress is then compared against F_y to determine whether the column falls in the elastic or inelastic buckling range.

#31 · Steel Structures

How do you determine whether a column is in the elastic or inelastic buckling range once F_e is known?

Answer

Compute the ratio F_y divided by F_e . If that ratio is less than or equal to 2.25, the column is in the inelastic range and $F_{cr} = 0.658$ raised to (F_y/F_e) times F_y . If it exceeds 2.25, the column is elastic and $F_{cr} = 0.877$ times F_e .

#32 · Steel Structures

A W12x53 column has $r_x = 5.23$ in and $r_y = 2.48$ in with equal unbraced lengths in both directions. Which axis governs slenderness and why?

Answer

The weak axis (r_y) governs because a smaller radius of gyration produces a larger KL/r for the same unbraced length. Weak axis $KL/r = 77.4$ versus strong axis $KL/r = 36.7$ in this example, so the weak axis controls.

#33 · Steel Structures

For the W12x53 example with $KL/r_y = 77.4$ and $F_y = 50$ ksi, what does F_e work out to, and which buckling range applies?

Answer

$F_e = \pi^2 \times 29000 \text{ divided by } 77.4^2$, which equals about 47.8 ksi. Since $F_y/F_e = 50/47.8 = 1.05$ is less than 2.25, the column is in the inelastic buckling range.

#34 · Steel Structures

For that same W12x53 column, what is F_{cr} and the final design axial strength ϕP_n ?

Answer

$F_{cr} = 0.658$ raised to 1.05 times 50, which is about 32.2 ksi. $P_n = F_{cr} \text{ times } A_g = 32.2 \text{ times } 15.6 = 502.3$ kips, and $\phi P_n = 0.90 \text{ times } 502.3 = 452.1$ kips.

#35 · Steel Structures

Where do candidates most often get the strong axis versus weak axis check backwards, and why?

Answer

Candidates instinctively check only the strong axis (r_x) because it feels like the obvious bending axis. In reality the weak axis (r_y) almost always has the larger KL/r and controls axial buckling, unless the member is braced differently in each direction, so both axes must always be calculated.

#36 · Steel Structures

Which AISC chapter governs flexural (beam) member design, and why does it consume the most exam time?

Answer

Beam design is AISC Chapter F. It consumes the most exam time because it branches into multiple paths depending on section compactness and unbraced length, unlike the more linear tension and simple compression checks.

#37 · Steel Structures

What is the first classification fork in flexural design, and where do the limits come from?

Answer

The first fork is classifying the section as compact, noncompact, or slender. This is done by comparing flange and web width-to-thickness ratios (b/t and h/t_w) against limits λ_{p-f} and λ_{r-f} in Table B4.1b.

#38 · Steel Structures

Are most common rolled W-shapes with $F_y = 50$ ksi compact, and what does that mean for the exam approach?

Answer

Yes, almost all common rolled W-shapes with $F_y = 50$ ksi are compact, so for most rolled shape problems you can go directly to Chapter F2 without a detailed compactness check. Even so, this should be quickly confirmed rather than blindly assumed, since built-up or slender web plate girders are typically not compact.

#39 · Steel Structures

For a compact section under Chapter F2, what governs strength when the unbraced length L_b is less than L_p ?

Answer

When L_b is less than L_p , the section reaches its full plastic moment, $M_n = M_p = F_y \text{ times } Z_x$. This is the maximum possible flexural strength since lateral torsional buckling does not limit it in this range.

#40 · Steel Structures

For a compact section, what governs strength when L_b falls between L_p and L_r ?

Answer

This zone is governed by inelastic lateral torsional buckling (LTB), where strength linearly interpolates downward from M_p at L_p toward a lower value at L_r . Candidates often get the direction of this trend backwards.

#41 · Steel Structures

What is the correct relationship between unbraced length and flexural capacity that candidates often reverse?

Answer

Shorter unbraced length gives more capacity, and longer unbraced length gives less capacity, down to a floor set by elastic lateral torsional buckling beyond L_r . Getting this backwards leads to picking the wrong trend when comparing bracing scenarios.

#42 · Steel Structures

What equation is used for inelastic LTB when L_p is less than L_b is less than L_r ?

Answer

$M_n = C_b \text{ times } [M_p \text{ minus } BF \text{ times } (L_b \text{ minus } L_p)]$, not to exceed M_p . BF is the tabulated slope factor for the section, and C_b accounts for the moment gradient along the unbraced segment.

#43 · Steel Structures

A W16x31 has $M_p = 225$ kip-ft, $L_p = 5.30$ ft, $L_r = 15.3$ ft, $BF = 4.85$ kips, and $L_b = 10$ ft with $C_b = 1.0$. What is M_n ?

Answer

Since L_p is less than L_b which is less than L_r , use inelastic LTB: $M_n = 1.0$ times $[225 \text{ minus } 4.85 \text{ times } (10 \text{ minus } 5.30)] = 225 \text{ minus } 22.8 = 202.2 \text{ kip-ft.}$

#44 · Steel Structures

For that W16x31 beam with $M_n = 202.2$ kip-ft, what is the LRFD design flexural strength ϕM_n ?

Answer

$\phi M_n = 0.90$ times $202.2 = 182.0 \text{ kip-ft.}$ This is the value compared against the factored required moment M_u for a pass/fail check.

#45 · Steel Structures

What happens to a beam's flexural capacity if it has continuous lateral bracing along its full length?

Answer

If L_b is less than L_p , the beam reaches $M_n = M_p$ with no LTB reduction at all, which is a much larger capacity than a beam with L_b between L_p and L_r . This is why checking bracing spacing is one of the first steps in any beam problem.

#46 · Steel Structures

What is the equation for shear strength of a rolled W-shape under AISC Chapter G?

Answer

$V_n = 0.6 \text{ times } F_y \text{ times } A_w \text{ times } C_v1$, where A_w is the web shear area. For most rolled shapes with h/t_w below the specified limit, $C_v1 = 1.0$.

#47 · Steel Structures

Does shear typically govern over flexure in ordinary rolled steel beams, and should it still be checked?

Answer

Shear rarely governs over flexure in typical gravity beams, but it can govern in short-span heavily loaded beams or at coped connections. It should never be skipped just because it usually does not control.

#48 · Steel Structures

Which AISC chapter governs members subject to combined axial load and bending, and what is this called?

Answer

This is beam-column interaction, checked under AISC Chapter H. The governing equations are H1-1a and H1-1b.

#49 · Steel Structures

What ratio determines which of the two Chapter H interaction equations to use?

Answer

The switch point is P_r divided by P_c equal to 0.2. Above that ratio, use equation H1-1a; at or below that ratio, use equation H1-1b.

#50 · Steel Structures

What is the form of interaction equation H1-1a, used when P_r/P_c exceeds 0.2?

Answer

H1-1a is P_r/P_c plus $(8/9)$ times $(M_{rx}/M_{cx}$ plus $M_{ry}/M_{cy})$, which must be less than or equal to 1.0. The $8/9$ factor reduces the weight given to the moment terms relative to the axial term.

#51 · Steel Structures

What is the form of interaction equation H1-1b, used when P_r/P_c is 0.2 or less?

Answer

H1-1b is $P_r/(2P_c)$ plus $(M_{rx}/M_{cx}$ plus $M_{ry}/M_{cy})$, which must be less than or equal to 1.0. Here the axial term is halved instead of being weighted at full value.

#52 · Steel Structures

What is the most common mistake candidates make on Chapter H interaction problems?

Answer

Candidates calculate P_r/P_c but forget to compare it against 0.2 before selecting an equation, applying the wrong interaction formula as a result. Always check which side of 0.2 the ratio falls on before picking H1-1a or H1-1b.

#53 · Steel Structures

A column has $P_r = 180$ kips, $P_c = 720$ kips. Which interaction equation applies?

Answer

$P_r/P_c = 180/720 = 0.25$, which is greater than 0.2, so equation H1-1a applies rather than H1-1b.

#54 · Steel Structures

For that column with $P_r/P_c = 0.25$, $M_{rx} = 95$ kip-ft, and $M_{cy} = 210$ kip-ft with no weak axis moment, does it pass H1-1a?

Answer

0.25 plus $(8/9)$ times $(95/210 \text{ plus } 0) = 0.25 \text{ plus } 0.402 = 0.652$, which is less than 1.0. The member passes the interaction check.

#55 · Steel Structures

Why is getting the Chapter H equation choice wrong especially costly on this exam?

Answer

Beam-columns in moment frames are one of the most commonly tested scenarios in the steel content. Using the wrong equation produces an incorrect pass/fail conclusion even if every individual capacity value was calculated correctly.

#56 · Steel Structures

Which AISC chapter governs composite steel-concrete members, and what is the key exam skill for this topic?

Answer

Composite beams are covered under AISC Chapter I. The key exam skill is locating the plastic neutral axis (PNA) and determining whether it falls within the slab or within the steel section.

#57 · Steel Structures

Why does the location of the plastic neutral axis matter for composite beam calculations?

Answer

The moment arm and stress block calculations change depending on whether the PNA lands in the slab or in the steel section. Getting this location wrong throws off the entire strength calculation.

#58 · Steel Structures

What three quantities compete to set the available compressive force in a composite beam's slab?

Answer

The available compressive force is the smallest of 0.85 times f'_c times A_c (concrete crushing), F_y times A_s (steel yielding), or the sum of stud shear capacities Q_n if the beam is only partially composite.

#59 · Steel Structures

What is partial composite action and why does the exam test it?

Answer

Partial composite action occurs when fewer shear studs are provided than needed for full composite behavior, which is common in real designs to save on studs. The exam tests whether candidates understand that partial composite beams have reduced strength AND reduced stiffness compared to full composite, affecting both strength and deflection checks.

#60 · Steel Structures

What does stud shear capacity per AISC I3.2d depend on?

Answer

Stud capacity depends on stud diameter, concrete strength, and deck configuration, including whether ribs run parallel or perpendicular to the beam and how many studs sit per rib.

#61 · Steel Structures

How does metal deck orientation affect stud capacity in a composite beam?

Answer

A metal deck oriented perpendicular to the beam with only one stud per rib reduces stud capacity compared to a solid slab. This detail is easy to miss when skimming a framing plan exhibit quickly.

#62 · Steel Structures

What are plate girders and what makes them different from typical rolled beams?

Answer

Plate girders are built-up sections, usually with slender webs whose h/t_w exceeds the compact or noncompact limits. They are governed by AISC Chapter G, Section G3 in addition to the base Chapter F and G equations.

#63 · Steel Structures

What is tension field action in a plate girder web?

Answer

Tension field action is a mechanism where a slender web carries additional shear beyond initial buckling by acting like a diagonal tension member spanning between stiffeners. It significantly increases post-buckling shear capacity.

#64 · Steel Structures

Under what condition can tension field action be applied to a plate girder web?

Answer

Tension field action only applies when transverse stiffeners are present at the proper spacing. A plate girder exhibit without stiffeners shown should not have tension field action applied to it.

#65 · Steel Structures

What unit confusion trap recurs across steel flexure problems on this exam?

Answer

L_p and L_r are typically given in feet, so L_b must also be compared in feet, but Z_x is in cubic inches, meaning M_p calculated as F_y times Z_x comes out in kip-inches. Convert that result to kip-feet by dividing by 12 before comparing against M_n tables already in kip-feet.

#66 · Steel Structures

What is the general habit that avoids unit confusion errors on multi-step steel calculations?

Answer

Confirm units at every single step rather than trusting that they will work out on their own. A single unmatched unit (kips versus pounds, inches versus feet) can silently make an otherwise correct calculation wrong.

#67 · Steel Structures

What is the difference between nominal strength and design strength, and why does confusing them matter?

Answer

Nominal strength (P_n , M_n) is the raw calculated capacity, while design strength is that value multiplied by ϕ (LRFD) or divided by ω (ASD). Handbook tables sometimes list nominal values and sometimes list design values directly, so misreading the table heading can produce an answer wrong by 10 to 65 percent depending on the limit state.

#68 · Steel Structures

Why is time management around the reference handbook itself part of steel exam strategy?

Answer

This is a closed-book exam with an open electronic reference, so time spent deriving an equation from scratch instead of navigating directly to the right table or figure is time you cannot recover. Candidates should practice locating specific tables quickly in the actual reference software before test day.

#69 · Steel Structures

Which three handbook locations does the chapter recommend practicing quick navigation to before test day?

Answer

Table D3.1 (shear lag factor U), Table B4.1b (compactness limits), and the Chapter H interaction equations are called out as high-value locations to be able to find quickly. Speed of lookup matters as much as knowing the concept exists.

#70 · Steel Structures

How are Vertical Buildings depth section problems typically structured on the exam?

Answer

Each problem presents a scenario tied to a building or structural exhibit, with several linked questions attached to that same set of given data. Candidates should read the whole scenario once and extract every given number into a clean list before answering.

#71 · Steel Structures

What common time-wasting mistake do candidates make with linked scenario questions?

Answer

Candidates often re-derive basic quantities like factored loads or section properties separately for each linked question instead of reusing values already calculated. Calculate shared quantities once and reuse them across every question in that scenario.

#72 · Steel Structures

Why should exam exhibits (framing plans, connection details, free body diagrams) always be scanned carefully before calculating?

Answer

Exhibits often carry information not repeated in the problem text, such as a stated bracing point, a labeled dimension, or a hole pattern. Assuming a standard condition when the exhibit shows something different is a common way to get a calculation wrong despite correct method.

#73 · Steel Structures

How do select-all (multiple-correct) items differ from typical single-answer multiple choice on this exam?

Answer

Select-all items do not tell you how many answers are correct, so you cannot assume exactly one right or one wrong choice out of habit. Every option must be evaluated independently against the code provisions and the scenario's specific numbers.

#74 · Steel Structures

What is the risk of rushing to pick the obvious answer on a select-all item?

Answer

Rushing to select what looks like the single obvious correct answer is one of the most common ways candidates lose points they otherwise knew how to earn. Each option must be checked on its own merits before deciding the combination.

#75 · Steel Structures

What special care is required for fill-in-the-blank numeric items on this exam?

Answer

These items require typing an exact value with no options to eliminate against for a sanity check. Carry extra decimal places through intermediate steps and round only at the final answer, since early rounding (like on a KL/r value used inside an exponent) can shift the result outside the grading tolerance.

#76 · Steel Structures

Besides rounding, what other error commonly ruins an otherwise correct fill-in-the-blank numeric answer?

Answer

Submitting the correct number in the wrong unit, such as kip-inches instead of kip-feet, will be marked wrong even though the underlying calculation was right. Always double check units before submitting a numeric entry.

#77 · Steel Structures

What is the actual skill the SE steel section is testing, according to the chapter's framing?

Answer

The exam is not testing memorization of every AISC 360 equation, since the reference handbook provides them all. It is testing judgment: knowing which chapter governs which failure mode, which table gives the shortcut, and how to apply numbers correctly under time pressure.

#78 · Steel Structures

If a W-shape tension member is bolted through both flanges only versus through both flange and web, does U change?

Answer

Yes. Connection configuration changes the shear lag factor U from Table D3.1, because U reflects how directly the load path reaches every part of the cross section, not just the shape category.

#79 · Steel Structures

For the tension member fracture limit state, what two effects does A_e actually combine?

Answer

A_e combines the physical loss of area from bolt holes (captured in A_n) and the shear lag reduction that accounts for non-uniform stress distribution across the section (captured by U). Both effects reduce the effective area below the gross area.

#80 · Steel Structures

Why might a slender column and a stocky column with the same A_g have very different P_n values?

Answer

The stocky column's F_{cr} approaches F_y since it fails by yielding, while the slender column's F_{cr} is limited by elastic buckling stress F_e , which decreases sharply as KL/r increases. $P_n = F_{cr}$ times A_g , so a lower F_{cr} directly reduces axial capacity even with identical cross section area.

#81 · Steel Structures

What is the boundary value of F_y/F_e that separates inelastic buckling from elastic buckling in AISC Chapter E?

Answer

A ratio of F_y/F_e less than or equal to 2.25 places the column in inelastic buckling, using $F_{cr} = 0.658$ raised to (F_y/F_e) times F_y . A ratio greater than 2.25 places it in elastic buckling, using $F_{cr} = 0.877$ times F_e .

#82 · Steel Structures

In the W16x31 flexure example, what would change if L_b were greater than L_r instead of between L_p and L_r ?

Answer

Beyond L_r , the beam enters elastic lateral torsional buckling, using a different equation based on F_e and the section properties rather than the linear inelastic interpolation between M_p and B_F times $(L_b \text{ minus } L_p)$. Capacity would continue dropping as L_b increases further.

#83 · Steel Structures

What role does the moment gradient factor C_b play in inelastic and elastic LTB equations?

Answer

C_b increases flexural capacity to account for a favorable moment gradient along the unbraced length, such as a beam with a lower moment at one brace point than the other. Using $C_b = 1.0$ is always conservative when the actual moment diagram is not uniform.

#84 · Steel Structures

Why would a built-up plate girder often fail to qualify for the straightforward Chapter F2 compact-section path?

Answer

Built-up plate girders commonly have slender webs with h/t_w exceeding the compact and noncompact limits in Table B4.1b, unlike common rolled W-shapes with $F_y = 50$ ksi which are almost always compact. This pushes plate girder design into Chapter G, Section G3 territory instead.

#85 · Steel Structures

How should a candidate handle a Chapter H interaction problem that also has a nonzero weak axis moment M_{ry} ?

Answer

Both M_{rx}/M_{cx} and M_{ry}/M_{cy} terms must be summed inside the moment portion of whichever equation applies, H1-1a or H1-1b. Ignoring the weak axis moment term understates the interaction ratio and can produce a false pass.

#86 · Steel Structures

Why is it risky to assume a composite beam is fully composite without checking the exhibit?

Answer

A framing plan or connection detail might show fewer shear studs than needed for full composite action, meaning the beam is actually partially composite with reduced strength and stiffness. Assuming full composite behavior when studs are limited overstates both strength and understates deflection.

#87 · Steel Structures

What is the difference in load path assumption between $0.85 f'_c A_c$ and $F_y A_s$ as limits on slab compressive force?

Answer

$0.85 f'_c A_c$ represents the concrete crushing in compression, limited by concrete area and strength. $F_y A_s$ represents the steel section yielding in tension, limited by steel area and strength. The smaller of the two, along with total stud capacity, controls the compressive force transferred.

#88 · Steel Structures

What common mistake could a candidate make by reading a nominal strength table value as if it were a design value?

Answer

Comparing a nominal P_n or M_n value directly against a factored required load without applying ϕ or ω overstates the actual usable capacity. Always read the table heading first to confirm whether it lists nominal or design (already-factored) values.

#89 · Steel Structures

Why does the exam prefer giving a stated K value or clear frame type instead of requiring a nomograph reading?

Answer

Reading an alignment chart (nomograph) from scratch for beam and column stiffness ratios is slow and imprecise under exam time pressure. The exam instead gives a stated K or a clear braced-frame versus moment-frame condition so candidates can move faster.

#90 · Steel Structures

If a scenario states a W18x50 beam with no grade given, what should you assume and why is that assumption safe?

Answer

Assume A992 with $F_y = 50$ ksi, since that is standard industry practice for W-shapes and exam writers rely on candidates knowing this default. Only deviate from this assumption if the scenario explicitly states a different grade.

#91 · Steel Structures

What is the practical exam impact of confusing yield strength F_y with allowable or design stress?

Answer

F_y is a material property used inside capacity equations, while allowable or design stress already reflects ϕ or ω applied to a calculated capacity. Treating F_y itself as a final allowable stress skips required steps and produces an incorrect capacity.

#92 · Steel Structures

How does an unbraced length L_b less than L_p differ physically from L_b greater than L_r in terms of failure mode?

Answer

When L_b is less than L_p , the beam yields fully in flexure before any lateral torsional buckling can occur. When L_b is greater than L_r , the beam buckles laterally and torsionally in the elastic range before reaching yield, at a reduced moment capacity.

#93 · Steel Structures

What single check should always be performed first on any beam problem before diving into Chapter F equations?

Answer

Check the bracing spacing (L_b) relative to L_p and L_r , since this determines which flexural strength regime applies and dramatically changes the resulting capacity. Skipping straight to a formula without this check risks applying the wrong equation.

#94 · Steel Structures

Why is the tension member fracture check said to depend on 'a smaller effective area with a lower phi factor'?

Answer

Fracture uses A_e , which is always smaller than or equal to A_g due to hole deductions and shear lag, combined with $\phi = 0.75$ which is lower than the yielding ϕ of 0.90. Both effects compound to often make fracture the governing, lower capacity even though F_u exceeds F_y .

#95 · Steel Structures

What should a candidate do if an exam scenario gives a beam span, load, and grade once but attaches five linked questions?

Answer

Calculate the factored moment, shear, and any other derived quantities a single time from the shared data, then reuse those same numbers across all five linked questions. Re-deriving them repeatedly wastes time without adding accuracy.

#96 · Steel Structures

Why does the chapter emphasize that judgment, not equation memorization, is the real skill being tested?

Answer

The NCEES electronic reference handbook already contains every AISC 360 equation used in these calculations. What the handbook cannot supply is deciding which chapter and equation applies to a given exhibit, which axis governs, or which limit state controls, and that decision-making is the actual skill under test.

#97 · Steel Structures

How should a candidate treat a plate girder exhibit that does not show any transverse stiffeners?

Answer

Tension field action should not be applied, since it requires stiffeners at proper spacing to develop the diagonal tension mechanism in the web. Without visible stiffeners in the exhibit, shear capacity should be based on simple web buckling behavior instead.

#98 · Steel Structures

What are the two competing limit states for a tension member, stated in one sentence?

Answer

A tension member can fail by yielding across its full gross section, or by fracturing across its reduced effective net area at a bolted or welded connection. Both must be checked and the lower governing value controls the design.

#99 · Steel Structures

Why is $E = 29000$ ksi used without further explanation in the F_e equation for steel columns?

Answer

29000 ksi is the standard modulus of elasticity for structural steel and is treated as a known constant on this exam, just like $F_y = 50$ ksi for A992. It should be used directly inside $F_e = \pi^2 E$ divided by $(KL/r)^2$ without needing to look it up each time.

#100 · Steel Structures

Summarize the fastest mental checklist for any steel beam-column combined loading problem on this exam.

Answer

First find governing axial capacity P_c and governing flexural capacity M_{cx} (and M_{cy} if present), then compute P_r/P_c and compare it to 0.2 to pick H1-1a or H1-1b, then plug in and compare the result to 1.0. Confirm units match at each step before finalizing the pass or fail conclusion.

#101 · Concrete Structures

What single material property drives most concrete design equations on the SE exam?

Answer

The concrete compressive strength, f'_c , drives almost every equation for flexure, shear, columns, and slabs. Typical exam values are 3,000 psi for slab-on-grade, 4,000 to 5,000 psi for beams, columns, and elevated slabs, and higher values for high-rise columns.

#102 · Concrete Structures

State the ACI 318-14 equation for modulus of elasticity of concrete, E_c .

Answer

$E_c = 33 \text{ times } w_c \text{ raised to the } 1.5 \text{ power times the square root of } f'_c$, with w_c in pounds per cubic foot and f'_c in psi. For normal weight concrete at 145 pcf this simplifies to the commonly used $E_c = 57,000 \text{ times the square root of } f'_c \text{ in psi}$.

#103 · Concrete Structures

What is the default reinforcing steel grade assumed on the SE exam unless stated otherwise?

Answer

Grade 60 rebar, with f_y equal to 60,000 psi, is the default assumption. Grade 40 and Grade 75 also appear on the exam, especially for older buildings or high seismic columns, so always read the given f_y rather than assuming 60 ksi automatically.

#104 · Concrete Structures

What is the area of a number 8 rebar?

Answer

A number 8 bar has a nominal diameter of 1.0 inch and an area of 0.79 square inches. The full bar area table for sizes 3 through 11 is available in the exam reference handbook, so only a few common sizes need to be recognized on sight.

#105 · Concrete Structures

List the areas of number 4, 5, 6, and 8 rebar from memory.

Answer

Number 4 has an area of 0.20 square inches, number 5 has 0.31 square inches, number 6 has 0.44 square inches, and number 8 has 0.79 square inches. Recognizing these on sight saves time since the exam handbook also has the full table available.

#106 · Concrete Structures

What design philosophy does ACI 318-14 use for strength calculations?

Answer

ACI 318-14 uses strength design, meaning factored loads are compared against a reduced nominal capacity. A strength reduction factor, ϕ , is applied to M_n , V_n , or P_n depending on the strain state of the section being checked.

#107 · Concrete Structures

What phi factor applies to tension-controlled flexure, and what value applies to shear?

Answer

Tension-controlled flexure uses phi equal to 0.90. Shear uses phi equal to 0.75. Mixing these two factors up is one of the most common careless mistakes candidates make on this exam.

#108 · Concrete Structures

What phi factor range applies to column design, and what governs where in that range a section falls?

Answer

Column phi ranges from 0.65 to 0.90 for tied columns, depending on whether the section is compression-controlled, in the transition zone, or tension-controlled. The governing factor is the net tensile strain in the extreme tension steel at nominal strength.

#109 · Concrete Structures

Why does the SE exam test judgment about phi factors rather than the numbers themselves?

Answer

The numeric values and equations are already sitting in the electronic reference handbook during the test. What is actually being tested is whether a candidate can identify which phi factor applies to a given scenario and execute the arithmetic correctly under time pressure.

#110 · Concrete Structures

Where in ACI 318-14 are flexural design provisions for beams and one-way slabs located?

Answer

Flexural design of reinforced concrete members is covered in ACI 318-14 Chapter 22. It rests on two core ideas, strain compatibility with the plane sections remain plane assumption, and an equivalent rectangular stress block that replaces the true nonlinear concrete stress distribution.

#111 · Concrete Structures

At what strain does concrete crush in ACI 318-14 flexural design?

Answer

Concrete is assumed to crush at an extreme compression fiber strain of 0.003. This value is fixed by the code and is the top of the strain compatibility triangle used to find neutral axis depth.

#112 · Concrete Structures

How is the stress block depth a related to the neutral axis depth c ?

Answer

The relationship is a equals β_1 times c , where β_1 is 0.85 for f'_c up to 4,000 psi. β_1 decreases by 0.05 for every 1,000 psi above 4,000 psi, down to a floor value of 0.65.

#113 · Concrete Structures

A beam has f'_c of 6,000 psi. What is β_1 ?

Answer

β_1 starts at 0.85 at 4,000 psi and drops 0.05 per 1,000 psi above that. At 6,000 psi that is a 2,000 psi increase, so β_1 equals 0.85 minus 0.10, which is 0.75.

#114 · Concrete Structures

Write the force equilibrium equation for stress block depth a in a singly reinforced rectangular beam.

Answer

a equals A_s times f_y divided by the quantity 0.85 times f'_c times b . This comes from setting the tension force in the steel equal to the compression force in the concrete stress block.

#115 · Concrete Structures

Write the nominal moment capacity equation for a singly reinforced rectangular beam.

Answer

M_n equals A_s times f_y times the quantity d minus a over 2. This represents the tension steel force multiplied by the lever arm between the tension steel and the centroid of the compression stress block.

#116 · Concrete Structures

What strain threshold defines a tension-controlled section in flexure?

Answer

A section is tension-controlled when the net tensile strain in the extreme tension steel, ϵ_t , equals or exceeds 0.005 at nominal strength. Tension-controlled sections get the maximum flexural ϕ of 0.90.

#117 · Concrete Structures

What strain threshold defines a compression-controlled section in flexure?

Answer

A section is compression-controlled when the net tensile strain in the extreme tension steel is at or below 0.002 at nominal strength. Between 0.002 and 0.005 the section is in a transition zone where ϕ varies linearly.

#118 · Concrete Structures

Why should a candidate verify ϵ_t on an exam beam problem instead of assuming tension-controlled behavior?

Answer

Most exam beam problems are designed to land in the tension-controlled zone, but a heavily over-reinforced beam can slip into the transition zone, lowering ϕ below 0.90. Assuming tension-controlled behavior without checking can silently produce a wrong design moment.

#119 · Concrete Structures

A rectangular beam has $b = 14$ in, $d = 21.5$ in, $f'_c = 4,000$ psi, $f_y = 60,000$ psi, and $A_s = 3.00$ in². Find a .

Answer

a equals A_s times f_y divided by 0.85 times f'_c times b , which is $180,000$ divided by $47,600$, giving 3.78 inches. This is the depth of the equivalent rectangular compression stress block.

#120 · Concrete Structures

For the beam in the prior card with $a = 3.78$ in, find the neutral axis depth c and confirm the strain classification.

Answer

c equals a divided by β_1 , which is 3.78 divided by 0.85 , giving 4.45 inches. ϵ_t equals 0.003 times d minus c over c , which works out to about 0.0115 , well above 0.005 , so the section is tension-controlled and ϕ equals 0.90 .

#121 · Concrete Structures

For that same beam, what is the final design moment ϕM_n ?

Answer

M_n equals A_s times f_y times d minus a over 2, which comes out to about 294.2 kip-feet. Multiplying by ϕ of 0.90 gives a design moment ϕM_n of about 264.8 kip-feet.

#122 · Concrete Structures

When is compression steel, A_s prime, added to a beam design?

Answer

Compression steel is added when a beam needs more capacity than the maximum single-layer tension-controlled limit allows, or when it is needed to help control long-term deflection and creep. It adds capacity beyond what the tension steel and stress block alone provide.

#123 · Concrete Structures

What check must be performed before assuming compression steel in a doubly reinforced beam has yielded?

Answer

The strain in the compression steel, found from the strain compatibility triangle, must be compared against the steel's yield strain, f_y divided by E_s . Skipping this check and assuming compression steel always yields is a common exam trap, especially on shallow sections with compression steel close to the neutral axis.

#124 · Concrete Structures

How are one-way slabs designed differently from beams?

Answer

One-way slabs are designed exactly the same way as beams, just on a per-foot-of-width basis using b equal to 12 inches. A single layer of bars at some spacing replaces the discrete bar count used in a beam.

#125 · Concrete Structures

What maximum bar spacing rule applies to one-way slabs for crack control under ACI 318-14?

Answer

ACI 318-14 Section 24.3 limits bar spacing to the smaller of 3 times the slab thickness or 18 inches. Recognizing that a slab check is really a spacing check rather than a strength check is a key exam skill.

#126 · Concrete Structures

Where does shear design for beams and one-way slabs live in ACI 318-14?

Answer

Shear design is covered in ACI 318-14 Chapter 22. The philosophy mirrors steel design, total nominal shear capacity is the sum of a concrete contribution V_c and a steel stirrup contribution V_s .

#127 · Concrete Structures

What phi factor applies to shear design, and what is the simplified V_c equation?

Answer

Phi equals 0.75 for shear. The simplified equation most candidates use is V_c equals 2 times lambda times the square root of f'_c times b_w times d , in pounds, with f'_c in psi and b_w , d in inches, and lambda equal to 1.0 for normal weight concrete.

#128 · Concrete Structures

What are the three shear zones used to decide stirrup requirements in a beam?

Answer

No stirrups are required when V_u is below ϕV_c over 2. Minimum stirrups are required between ϕV_c over 2 and ϕV_c . Designed stirrups are required above ϕV_c , where V_s equals V_u over ϕ minus V_c .

#129 · Concrete Structures

Write the equation for required stirrup spacing once V_s is known.

Answer

s equals A_v times f_y times d divided by V_s . This spacing must then be checked against the code's maximum spacing limits before being accepted as the final answer.

#130 · Concrete Structures

What are the maximum stirrup spacing limits in ACI 318-14, and what triggers the tighter limit?

Answer

The general maximum spacing is d over 2, tightening to d over 4 when V_s exceeds 4 times the square root of f'_c times b_w times d . An absolute cap of 24 inches applies in the d over 2 regime, but a tighter absolute cap of 12 inches applies in the d over 4 regime.

#131 · Concrete Structures

A beam has $b_w = 12$ in, $d = 20$ in, $f'_c = 4,000$ psi. Find V_c using the simplified equation.

Answer

V_c equals 2 times 1.0 times the square root of 4,000 times 12 times 20, which is 2 times 63.25 times 240, giving 30,360 pounds or 30.36 kips. This is the concrete's nominal shear contribution before any stirrups are added.

#132 · Concrete Structures

For that beam with $V_c = 30.36$ kips and $V_u = 42$ kips, is stirrup design required by strength?

Answer

ϕV_c equals 0.75 times 30.36, which is 22.77 kips. Since V_u of 42 kips exceeds ϕV_c , stirrups are required by strength, not merely by the minimum reinforcement rule.

#133 · Concrete Structures

For that same beam, find the required V_s and the resulting stirrup spacing using number 3 double-leg stirrups, $A_v = 0.22 \text{ in}^2$.

Answer

V_s required equals V_u over ϕ minus V_c , which is 56.0 minus 30.36, giving 25.64 kips. Spacing s equals A_v times f_y times d divided by V_s , which comes out to about 10.3 inches.

#134 · Concrete Structures

In that same stirrup problem, does the calculated 10.3 inch spacing actually govern the final answer?

Answer

No. Since V_s of 25.64 kips is below the 60.7 kip threshold, the spacing cap is d over 2, which equals 10 inches. Because the calculated 10.3 inches exceeds that 10 inch cap, the final spacing must be reduced to 10 inches, governed by the maximum spacing limit rather than the strength calculation.

#135 · Concrete Structures

What is a commonly missed step in shear stirrup design problems on this exam?

Answer

Many candidates calculate a spacing from the strength equation and stop there, without comparing it against the code maximum spacing limit and selecting whichever value is smaller. The handbook lists every limit but will not remind a candidate to actually apply it.

#136 · Concrete Structures

What is the interaction diagram in column design, and what does it plot?

Answer

The interaction diagram plots nominal axial capacity P_n against nominal moment capacity M_n for a specific cross section and reinforcement layout. Every point on the curve represents a different neutral axis location for that section.

#137 · Concrete Structures

What phi factor applies at the top of the interaction diagram for a tied column versus a spiral column?

Answer

At pure concentric axial load, tied columns use phi equal to 0.65 and spiral columns use phi equal to 0.75. This 0.10 difference directly rewards the extra confinement that spiral reinforcement provides.

#138 · Concrete Structures

What happens to the column phi factor as the section moves toward pure flexure at the bottom of the interaction diagram?

Answer

Phi transitions upward from its compression-controlled starting value toward 0.90 as the strain state shifts into the tension-controlled zone. At the very bottom of the curve the section behaves essentially like a beam.

#139 · Concrete Structures

Do candidates need to derive a column's interaction diagram from scratch on the SE exam?

Answer

No. The reference handbook supplies pre-built interaction diagrams for common reinforcement ratios and cover conditions. The actual exam skill is reading the correct chart accurately, not deriving the curve mathematically.

#140 · Concrete Structures

What is the maximum axial capacity equation for a tied column, including the code-required eccentricity reduction?

Answer

$P_n(\max)$ equals 0.80 times the quantity 0.85 times f'_c times A_g minus A_{st} , plus f_y times A_{st} . The 0.80 factor accounts for the minimum eccentricity that ACI 318-14 requires tied columns to be designed for.

#141 · Concrete Structures

A 16 in square tied column has $f'_c = 5,000$ psi, $f_y = 60,000$ psi, $A_{st} = 8.00$ in², $A_g = 256$ in². Find $P_n(\max)$.

Answer

$P_n(\max)$ equals 0.80 times the quantity 0.85 times 5,000 times 248, plus 60,000 times 8.00, which is 0.80 times 1,534,000, giving 1,227,200 pounds or 1,227.2 kips. This is the top point of that column's interaction diagram before ϕ is applied.

#142 · Concrete Structures

For that same 16 in square tied column with $P_n(\max) = 1,227.2$ kips, find the design axial capacity $\phi P_n(\max)$.

Answer

ϕ equals 0.65 for a tied column at pure axial load, so $\phi P_n(\max)$ equals 0.65 times 1,227.2, giving 797.7 kips. Any combination of P_u and M_u falling inside the resulting interaction curve is an acceptable design.

#143 · Concrete Structures

What triggers the need to check slenderness effects in a column?

Answer

Slenderness matters once a column's unsupported length becomes long relative to its cross-sectional dimension, measured through the kl_u/r ratio. The exam usually signals this by giving an unusually long unbraced height or a slim cross section relative to that height.

#144 · Concrete Structures

What is the approximate slenderness limit below which a braced column can be treated as short?

Answer

For braced frames, a column can be treated as short when kl_u/r is below roughly $34 - 12 \frac{M_1}{M_2}$. Below this limit slenderness effects can be ignored in the design.

#145 · Concrete Structures

What is the approximate slenderness limit for an unbraced (sway) column?

Answer

For unbraced or sway frames, a column can be treated as short when kl_u/r is below roughly 22. This limit is stricter than the braced frame case because sway frames are more sensitive to second-order effects.

#146 · Concrete Structures

What must be done if a column fails the short-column slenderness check?

Answer

The design moment must be magnified, using either the ACI moment magnifier method or a full second-order P-delta analysis. Ignoring a failed slenderness check and using the unmagnified moment understates the true demand on the column.

#147 · Concrete Structures

Which ACI 318-14 chapter governs two-way slab systems like flat plates and flat slabs?

Answer

Two-way slab systems fall under ACI 318-14 Chapter 8 for the direct design method and equivalent frame method analysis. Punching shear for these same systems is checked separately under Chapter 22.

#148 · Concrete Structures

Write the direct design method equation for total static moment, M_o , in a two-way slab panel.

Answer

M_o equals w_u times l_2 times l_n squared, all divided by 8. This total static moment is then distributed into column strip and middle strip moments using fixed percentage tables in the handbook.

#149 · Concrete Structures

What determines the column strip and middle strip moment percentages in the direct design method?

Answer

The percentages depend on span ratios and whether the panel is an interior span or an exterior span. These percentage tables are provided in the reference handbook, so the exam skill is correctly identifying panel type and strip, not memorizing the percentages.

#150 · Concrete Structures

Why is punching shear considered one of the highest-yield topics on the SE exam for slab systems?

Answer

Punching shear, or two-way shear, is the failure mode unique to slab systems and it governs the design of many flat plate structures. Because it appears so frequently and controls so many designs, it is heavily tested.

#151 · Concrete Structures

Where is the critical section for punching shear located relative to the column?

Answer

The critical section is located at a distance of d over 2 from the face of the column. It forms a shape that follows the perimeter of the column, offset outward by that d over 2 distance on every side.

#152 · Concrete Structures

What three variables govern which of the three punching shear equations controls in ACI 318-14 Chapter 22?

Answer

The three governing variables are β , the column's long-to-short side aspect ratio, α_s , a location factor, and b_o , the critical shear perimeter. The smallest of the three resulting V_c values must be used.

#153 · Concrete Structures

What α_s values apply to interior, edge, and corner columns in punching shear design?

Answer

α_s equals 40 for interior columns, 30 for edge columns, and 20 for corner columns. A lower α_s value reduces punching shear capacity, so edge and corner columns are more vulnerable to this failure mode.

#154 · Concrete Structures

An interior 18 in square column supports a flat plate with $d = 7$ in. Find the critical perimeter b_o .

Answer

The critical section is offset d over 2, or 3.5 inches, from each face, making the critical section a square with side length 18 plus 3.5 plus 3.5, equal to 25 inches. b_o equals 4 times 25, giving 100 inches.

#155 · Concrete Structures

For that same interior column with $b_o = 100$ in, $d = 7$ in, $f'_c = 4,000$ psi, find V_c using the simplest of the three punching shear equations.

Answer

For an interior column with equal column dimensions, β equals 1.0, so the simple equation V_c equals 4 times λ times the square root of f'_c times b_o times d applies. This gives V_c equal to 4 times 1.0 times 63.25 times 100 times 7, or 177,100 pounds, which is 177.1 kips.

#156 · Concrete Structures

For that same slab-column connection, find ϕV_c .

Answer

ϕ equals 0.75 for shear, so ϕV_c equals 0.75 times 177.1 kips, giving 132.8 kips. This is the design punching shear capacity to compare against the factored shear demand at that column.

#157 · Concrete Structures

Why must all three punching shear equations be calculated rather than just the simplest one?

Answer

A slab with an unusual column aspect ratio, or an edge or corner column, can fail the beta-based or α_s -based equations even when the simple flat equation would have passed. The smallest of the three results always governs.

#158 · Concrete Structures

Where in ACI 318-14 is development length covered?

Answer

Development length, the embedment length a bar needs to transfer its full tensile force into the surrounding concrete through bond, is covered in ACI 318-14 Chapter 25. The basic tension development length equation factors in bar size, f_y , f'_c , bar location, and coating.

#159 · Concrete Structures

What is the top-bar effect in development length calculations?

Answer

Top bars, meaning horizontal reinforcement with more than 12 inches of fresh concrete cast beneath them, require a longer development length. This is because trapped bleed water and air voids weaken the bond beneath top-cast bars compared to bottom bars.

#160 · Concrete Structures

How does epoxy coating affect required development length?

Answer

Epoxy coating reduces bond strength between the bar and surrounding concrete, so epoxy-coated bars require a longer development length than uncoated bars of the same size in the same condition. This is applied as a modification factor to the base development length equation.

#161 · Concrete Structures

Rank straight tension development length, straight compression development length, and standard hook development length in tension from longest to shortest.

Answer

Straight tension development length is always the longest of the three for the same bar. Straight compression development length is shorter, since compression bars do not rely on the same bond mechanism through cracked tension zones. Standard 90-degree hook development length in tension is shorter yet, since the hook provides direct mechanical anchorage.

#162 · Concrete Structures

Why is a hooked bar development length shorter than a straight bar tension development length?

Answer

A hook provides direct mechanical anchorage at the end of the bar, in addition to whatever bond develops along its length. This mechanical anchorage lets the bar transfer its full force in a shorter embedment than a straight bar relying on bond alone.

#163 · Concrete Structures

What is a common exam trap involving hooked bars at a support?

Answer

Candidates sometimes apply a straight tension development length equation when the exhibit actually shows a hooked bar terminating at a support. This silently overstates the required length and leads to the wrong answer, so always check whether the detail shown is a hook or a straight bar.

#164 · Concrete Structures

What is the difference between a lap splice and a mechanical or welded splice?

Answer

A lap splice overlaps two bars and transfers force through the surrounding concrete over a specified lap length. A mechanical or welded splice does not depend on concrete bond at all, instead transferring force through a direct connector or weld.

#165 · Concrete Structures

What is the difference between a Class A and a Class B lap splice length?

Answer

A Class A splice requires a lap length of 1.0 times the development length l_d . A Class B splice requires 1.3 times l_d , which is 30 percent more length, and is required when more than half the bars at a section are spliced at the same location or the splice is at a point of maximum stress.

#166 · Concrete Structures

What bar sizes cannot use lap splices under ACI 318-14?

Answer

Bar sizes larger than number 11 cannot use lap splices at all and must use mechanical or welded connections instead. This is a hard code limit rather than a calculated value, so the exam likes to test it directly.

#167 · Concrete Structures

What is the single biggest concrete exam trap regarding phi factors?

Answer

The single biggest trap is mixing up phi factors between flexure at 0.90, shear at 0.75, and columns which range from 0.65 to 0.90 depending on strain. Always confirm which failure mode is being checked before applying a phi value.

#168 · Concrete Structures

What two reinforcement limits must every beam design satisfy, beyond just meeting the required moment capacity?

Answer

A beam must satisfy a minimum reinforcement floor, As minimum, tied to the square root of f'_c over f_y and to 200 over f_y , and it must also effectively satisfy a ceiling through the tension-controlled strain requirement. A beam that fails either check is not a valid design even if the raw moment capacity number looks acceptable.

#169 · Concrete Structures

Why do concrete problems on the SE exam tend to have more steps than steel problems?

Answer

Concrete design almost always involves checking two materials at once, concrete in compression and steel in tension, plus a strain compatibility assumption tying them together. Steel problems typically involve fewer simultaneous material checks.

#170 · Concrete Structures

What is the actual skill being tested on the concrete portion of the SE exam, given that equations are in the handbook?

Answer

The exam tests whether a candidate can scan a scenario and exhibit, correctly identify which topic area the problem belongs to, pull the right equation from the handbook quickly, and execute the arithmetic without a sign error or unit mismatch. It is not testing memorization of the equations themselves.

#171 · Concrete Structures

What strain compatibility assumption underlies all reinforced concrete flexural design?

Answer

The plane sections remain plane assumption, meaning strain varies linearly across the depth of a cross section. This assumption ties together the concrete compressive strain and the steel tensile strain at any given neutral axis location.

#172 · Concrete Structures

In the tension-controlled beam example with $A_s = 3.00 \text{ in}^2$ and $M_n = 294.2 \text{ kip-ft}$, what would happen to ϕM_n if the section were instead in the transition zone?

Answer

ϕ would fall somewhere between 0.65 and 0.90 instead of the maximum 0.90, linearly interpolated based on ϵ_t . A lower ϕ directly reduces the design moment ϕM_n even though M_n itself stays the same.

#173 · Concrete Structures

Why is 3,000 psi concrete a realistic exam value for slab-on-grade but not for an elevated beam?

Answer

Slab-on-grade sits directly on soil and typically carries lighter, more distributed loads with less concern for punching or high flexural demand. Elevated beams, columns, and slabs need higher strength, typically 4,000 to 5,000 psi, to handle greater structural demands efficiently.

#174 · Concrete Structures

What happens to β_1 once f'_c reaches very high strengths, such as 10,000 psi?

Answer

β_1 decreases by 0.05 for every 1,000 psi above 4,000 psi but is floored at 0.65, so it cannot decrease further no matter how high f'_c climbs. At f'_c around 8,000 psi and above, β_1 stays fixed at 0.65.

#175 · Concrete Structures

In the punching shear worked example, what would change if the column were a corner column instead of interior?

Answer

α_s would drop from 40 to 20, which lowers the α_s -based V_c equation substantially. The α_s -based equation could then become the smallest and governing equation, even though it looked unimportant in the interior column case.

#176 · Concrete Structures

What is the role of λ in the concrete shear and punching shear equations?

Answer

λ is a modification factor for lightweight concrete, equal to 1.0 for normal weight concrete and less than 1.0 for lightweight concrete. It reduces V_c because lightweight aggregate concrete generally has lower shear friction and bond characteristics than normal weight concrete.

#177 · Concrete Structures

In the doubly reinforced beam scenario, what physically happens if compression steel has NOT actually yielded but a candidate assumes it has?

Answer

Assuming yield when the compression steel strain is actually below f_y over E_s overstates the compression steel's contributing stress, since real stress in that steel would be less than f_y . This overstates M_n and can produce an unconservative, incorrect design moment.

#178 · Concrete Structures

Why does a spiral column get a higher starting phi factor than a tied column?

Answer

Spiral reinforcement provides continuous confinement around the core concrete, giving more ductile behavior and a more gradual failure than the sudden spalling seen in tied columns. ACI 318-14 rewards this improved behavior with phi starting at 0.75 instead of 0.65 in the compression-controlled zone.

#179 · Concrete Structures

What does it mean for a beam design to be over-reinforced in the context of phi factors?

Answer

An over-reinforced beam has more tension steel than needed to reach the tension-controlled strain limit, so the neutral axis sits deeper and ϵ_t falls below 0.005. This pushes the section into the transition zone or compression-controlled zone, reducing ϕ below 0.90.

#180 · Concrete Structures

What is the physical meaning of the term d minus a over 2 in the beam moment capacity equation?

Answer

It represents the internal lever arm, the distance between the centroid of the tension steel force and the centroid of the compression stress block. Multiplying this lever arm by the internal force gives the internal resisting moment, M_n .

#181 · Concrete Structures

Why does a slender column need moment magnification instead of just using the first-order analysis moment?

Answer

As a slender column deflects laterally under axial load, the axial load acting through that lateral deflection creates additional moment, known as the P-delta effect. A first-order analysis ignores this added moment, so it must be magnified to represent the true demand.

#182 · Concrete Structures

What is the practical difference between the ACI moment magnifier method and a second-order P-delta analysis?

Answer

The moment magnifier method uses a simplified multiplier applied to the first-order moment to approximate second-order effects by hand. A full second-order P-delta analysis directly accounts for the added moment through iterative or computer analysis rather than an approximate multiplier.

#183 · Concrete Structures

Why does the exam consider a slim column with a long unbraced height a signal for slenderness checks?

Answer

The radius of gyration, r , is small for a slim cross section, and a long unbraced height, l_u , increases the kl_u/r ratio. A large kl_u/r ratio is exactly what triggers the need to check whether the column can be treated as short or must be magnified for slenderness.

#184 · Concrete Structures

In two-way slab design, what does the equivalent frame method offer compared to the direct design method?

Answer

Both methods are permitted under ACI 318-14 Chapter 8 for analyzing two-way slabs, but the equivalent frame method models the slab-column system as a frame and can handle irregular geometry or loading that the direct design method's simpler percentage-based approach cannot.

#185 · Concrete Structures

Why is punching shear checked separately from one-way (beam action) shear in a flat plate?

Answer

Punching shear is a two-way failure mode where the column tends to punch through the slab along a perimeter shear surface, unlike one-way shear which acts across a straight section. Because the failure geometry is fundamentally different, ACI 318-14 provides distinct equations for it.

#186 · Concrete Structures

What is the significance of the $200/f_y$ term in the minimum reinforcement equation for beams?

Answer

ACI 318-14 requires A_s minimum to be the larger of a term based on the square root of f'_c over f_y and the term $200/f_y$. The $200/f_y$ term acts as a floor that becomes controlling for lower strength concrete, ensuring a beam does not fail suddenly and brittlely right after first cracking.

#187 · Concrete Structures

Why does the exam emphasize checking units and sign conventions on concrete problems?

Answer

Concrete equations frequently mix pounds and kips, and inches and feet, across a single calculation, such as V_c in pounds converting to kips for comparison with V_u given in kips. A units mismatch produces an answer that is wrong by a factor of 1,000, a very common and avoidable exam error.

#188 · Concrete Structures

A beam has V_s calculated at 65 kips with $b_w = 12$ in, $d = 20$ in, $f'_c = 4,000$ psi. Does the tighter $d/4$ spacing cap apply?

Answer

The tighter cap of $d/4$ applies when V_s exceeds 4 times the square root of f'_c times b_w times d , which was calculated as 60.7 kips in the worked example conditions. Since 65 kips exceeds 60.7 kips, the maximum spacing must be reduced to d over 4 instead of d over 2.

#189 · Concrete Structures

What does it mean that the exam handbook contains modification factor tables for development length?

Answer

It means the candidate does not need to memorize numeric coefficients for bar location, coating, or confinement effects. The exam skill is correctly identifying which modification factors apply to the specific bar and detail shown, then applying them from the table.

#190 · Concrete Structures

Why is recognizing panel type, interior versus exterior, important in the direct design method?

Answer

The percentage of total static moment M_o assigned to the column strip and middle strip differs depending on whether the panel is an interior span or an exterior span. Misidentifying panel type applies the wrong percentage split, producing incorrect column strip and middle strip design moments.

#191 · Concrete Structures

In the tied column axial capacity equation, why is the term A_g minus A_{st} used rather than just A_g ?

Answer

The concrete stress term, 0.85 times f'_c , only applies to the net concrete area, which is the gross area A_g minus the area displaced by the longitudinal steel, A_{st} . The steel term, f_y times A_{st} , separately accounts for the force carried by the reinforcing bars themselves.

#192 · Concrete Structures

What is the physical reason the code applies a 0.80 reduction factor to $P_n(\max)$ for tied columns?

Answer

ACI 318-14 requires tied columns to be designed for a minimum eccentricity even under nominally concentric load, since perfectly centered loading rarely occurs in practice. The 0.80 factor accounts for this required minimum eccentricity by capping the usable pure axial capacity below the theoretical maximum.

#193 · Concrete Structures

Why might an exam problem describe a beam-column joint at the edge of a building as needing a hooked bar detail?

Answer

At an edge of a building there may not be enough straight embedment length available past the joint to develop the bar in tension using a straight bar. A hook lets the bar develop its full force in a shorter length through mechanical anchorage, which fits the limited space at that edge condition.

#194 · Concrete Structures

What is the difference in reasoning between checking maximum reinforcement in a beam versus checking minimum reinforcement?

Answer

Minimum reinforcement prevents a sudden brittle failure right at first cracking by ensuring enough steel is present to carry the cracking moment. Maximum reinforcement, enforced through the tension-controlled strain requirement, ensures the beam fails in a ductile manner with yielding steel rather than sudden concrete crushing.

#195 · Concrete Structures

Why is it important to double check whether a punching shear column is square versus rectangular before selecting an equation?

Answer

The beta term, the ratio of the column's long side to short side, equals 1.0 only for a square column and increases for rectangular columns, which can lower the beta-based V_c equation below the simple flat equation. A rectangular column can shift which of the three equations governs compared to a square column of similar size.

#196 · Concrete Structures

What is the takeaway if a calculated stirrup spacing comes out larger than the maximum allowed spacing?

Answer

The final design spacing must be the maximum allowed spacing, not the larger calculated value, since a wider spacing than allowed would violate the code limit even though it satisfies the strength equation. Always select the smaller of the calculated spacing and the code maximum.

#197 · Concrete Structures

Why does epoxy coating combined with the top-bar effect on the same bar multiply rather than simply add together?

Answer

ACI 318-14 applies bar location and coating as separate multiplicative modification factors to the base development length, not as additive corrections. A top bar that is also epoxy-coated gets both factors multiplied into the base l_d , compounding the required length increase.

#198 · Concrete Structures

What is the key difference in required lap length between a Class A splice and a Class B splice, and which is the more commonly required class in practice?

Answer

Class A requires 1.0 times l_d while Class B requires 1.3 times l_d , 30 percent more length. Class B is triggered whenever more than half the bars at a section are spliced at the same location or the splice sits at a point of maximum stress, which makes Class B the more commonly governing condition in real designs.

#199 · Concrete Structures

Summarize why the exam frames concrete design around seven distinct problem categories rather than one unified topic.

Answer

Concrete design spans materials, flexure, shear, columns, two-way slabs, and development length and splices, each with its own governing chapter and equation set in ACI 318-14. Fast, accurate exam performance depends on correctly sorting a given scenario into the right one of these categories before pulling any equation.

#200 · Concrete Structures

What is the final habit the chapter recommends after finishing any concrete calculation on exam day?

Answer

Always ask whether the resulting number makes physical sense for the size and type of member being checked. Combined with double checking phi factors and units, this final sanity check catches many of the errors that a purely mechanical calculation would miss.

#201 · Wood Structures

On the SE exam, should you memorize F_b , F_t , F_v , and F_c for common species and grades?

Answer

No. The NCEES electronic reference handbook gives you the actual NDS Supplement tables during the exam. Your job is to find the right table fast, not to recall a tabulated number from memory. What you must know cold is which adjustment factors apply and why.

#202 · Wood Structures

What does F_c perpendicular represent, and where does it get used?

Answer

F_c perpendicular is compression perpendicular to grain, which physically means bearing stress. It governs the bearing check where a beam rests on a support, not the axial compression check on a column. Confusing this with F_c parallel to grain is a common time-pressure mistake.

#203 · Wood Structures

What is CD, and what determines its value?

Answer

CD is the load duration factor. Its value is set by the shortest duration load in the load combination being checked, not by the load with the largest magnitude. Dead plus live uses 1.0, a combination including snow uses 1.15, and wind or seismic combinations use 1.6.

#204 · Wood Structures

Trap: a beam is checked under D plus L plus S plus W. A candidate sets CD based on the largest load in the combination. What is wrong?

Answer

CD depends on duration, not magnitude. Even if snow is the biggest load numerically, if wind is part of that combination the shortest duration load present is wind, so CD equals 1.6 for that check. Picking CD from the biggest number instead of the shortest duration load gives a wrong answer.

#205 · Wood Structures

When does CM, the wet service factor, apply?

Answer

CM applies when the wood will experience sustained in service moisture content above 19 percent. Most interior framing in a typical building stays dry and does not get this factor. Applying CM reflexively to an interior floor joist wastes capacity for no physical reason.

#206 · Wood Structures

What is Ct and when is it relevant on a typical building exam question?

Answer

Ct is the temperature factor, which reduces design values near sustained high heat sources. It is rarely relevant on typical building depth exam problems since most structures do not involve sustained elevated temperatures. Like CM, do not apply it unless the scenario actually describes the exposure.

#207 · Wood Structures

What physical phenomenon does CL, the beam stability factor, address?

Answer

CL accounts for lateral torsional buckling of a bending member whose compression edge is not continuously braced. It is analogous in concept to lateral torsional buckling checks in steel beam design, just with a different closed form NDS equation.

#208 · Wood Structures

What is CF, and why can it not be treated as a single number for a given member?

Answer

CF is the size factor, correcting tabulated values because small reference test specimens do not perfectly predict full size behavior. It is tabulated separately for each property, so the CF used for F_b is not necessarily the same value used for F_t or F_c on the same member.

#209 · Wood Structures

A 6x6 post needs a size factor for F_c . Can you use the dimension lumber CF value from memory?

Answer

No. Posts and timbers, five inches by five inches and larger, are a separate size category from dimension lumber with their own baseline assumptions. You must check which table category the exhibit's actual member falls into rather than carrying over a dimension lumber CF.

#210 · Wood Structures

What is C_i , and which design values does it affect?

Answer

C_i is the incising factor, applied when lumber is mechanically incised for preservative treatment. It reduces multiple properties at once, including E , F_b , F_t , F_c , and F_v , since the incisions cut fibers throughout the cross section.

#211 · Wood Structures

What is C_r , and what physical conditions must be met to use it?

Answer

C_r is the repetitive member factor, a 1.15 increase applied only to F_b . It requires three or more parallel members spaced no more than 24 inches on center and connected by a load distributing element like sheathing, which describes typical floor and roof joist framing.

#212 · Wood Structures

Trap: a single isolated beam that does not share sheathing with adjacent parallel members is given C_r equal to 1.15. Why is this wrong?

Answer

C_r requires a repetitive system of three or more parallel members at 24 inches on center or less, tied together by a load distributing element. A single isolated beam does not meet that condition, so applying C_r overstates its bending capacity.

#213 · Wood Structures

Which two NDS chapters govern sawn lumber reference design values and adjustment factors?

Answer

NDS Chapter 4 together with the NDS Supplement give the tabulated reference design values for sawn lumber. The adjustment factor definitions and applicability, such as CD, CM, CL, CF, Ci, and Cr, are also set out in this same reference framework before you touch beam or column design in Chapter 3.

#214 · Wood Structures

What three limit states, plus one additional check, does sawn lumber beam design under NDS Chapter 3 address?

Answer

Beam design checks bending stress, shear stress, and deflection as the three primary limit states. In addition, a bearing check at the supports is required to confirm the wood can resist the perpendicular to grain compression from the reaction.

#215 · Wood Structures

Write the bending stress check for a sawn lumber beam.

Answer

Bending stress f_b equals M divided by S , the section modulus. This computed f_b is compared against F_b prime, the reference F_b multiplied by all applicable adjustment factors such as C_D , C_L , C_F , and C_r where relevant.

#216 · Wood Structures

Why is the shear stress formula for a rectangular wood beam f_v equals $1.5V$ over (bd) , instead of simply V over the area?

Answer

The shear stress distribution across a rectangular cross section is parabolic, with the peak occurring at the neutral axis, not uniform. A simple average V over A underestimates that true peak stress, so the 1.5 factor corrects for the actual parabolic distribution.

#217 · Wood Structures

What does NDS 3.4.3 allow regarding the location of the design shear check, and is it mandatory?

Answer

NDS 3.4.3 permits computing the design shear at a distance d from the face of the support rather than at the support itself, since concentrated reactions near a support do not fully load the beam in shear the way loads farther along the span do. This reduction is optional, not mandatory, and a conservative engineer can instead check shear at the full reaction.

#218 · Wood Structures

What governs the deflection limit used for a wood beam, and where does that limit come from?

Answer

The deflection limit, such as L over 360 for live load or L over 240 for total load, is set by the applicable building code or the project's own design criteria. It is not a fixed NDS number, so a specific limit must be given or referenced in the problem, not assumed from memory.

#219 · Wood Structures

For a simple span uniformly loaded wood beam, what is the deflection equation, and which E value goes into it?

Answer

Deflection equals $5wL^4$ to the fourth divided by $384EI$, the standard mechanics of materials formula. The E used is the tabulated design modulus of elasticity, not E_{min} , because deflection is a serviceability check rather than a stability check.

#220 · Wood Structures

What is E_{min} , and when is it used instead of the tabulated E?

Answer

E_{min} is a reduced modulus of elasticity used specifically for stability related checks, such as beam lateral torsional buckling (CL) and column buckling (Cp), not for serviceability deflection calculations. Confusing E and E_{min} between a stability check and a deflection check produces an inconsistent design.

#221 · Wood Structures

What factor primarily adjusts the bearing check at a beam support for short bearing lengths?

Answer

C_b , the bearing area factor, applies for short bearing lengths near a beam end and can increase the effective F_c perpendicular capacity. C_M and C_t can also apply to the bearing check when the physical conditions for wet service or elevated temperature are actually present.

#222 · Wood Structures

What NDS chapter, beyond Chapter 3's beam provisions, covers wood column design and its central buckling concept?

Answer

NDS Chapter 3 also covers columns, and the central concept is the column stability factor C_p , which accounts for buckling in a manner conceptually similar to the χ factor in steel design, though computed with a different closed form equation specific to wood.

#223 · Wood Structures

Write the equation for F_{cE} , the critical buckling stress used in wood column design.

Answer

F_{cE} equals 0.822 times E_{min} prime divided by the quantity $(L_e \text{ over } d)$ squared. This is computed separately for each axis of the column using the actual unbraced length and section dimension appropriate to that axis.

#224 · Wood Structures

Which axis of a wood column generally governs the F_{cE} calculation, and why?

Answer

The axis with the larger $L_e \text{ over } d$ ratio generally governs because a larger slenderness ratio produces a lower, more critical F_{cE} value. You must compare both axes rather than assuming the visually shorter or weaker looking direction automatically controls.

#225 · Wood Structures

What is the NDS 3.7.1.4 slenderness limit for sawn lumber columns, and what does exceeding it mean?

Answer

NDS 3.7.1.4 caps L_e over d at 50 for sawn lumber columns. An exhibit showing an unbraced length that produces a ratio beyond 50 means the member fails outright by definition, regardless of what the computed stress ratio would otherwise show.

#226 · Wood Structures

In the NDS 3.7.1 column stability equation, what is F_c star, and how does it differ from F_c prime?

Answer

F_c star is the reference F_c adjusted by every applicable factor except C_p itself. F_c prime is the final adjusted design value, obtained by multiplying F_c star by C_p , and it is F_c prime that gets compared against the actual axial stress P over A .

#227 · Wood Structures

What are the three values of the column parameter c used in the C_p equation, and which member types do they apply to?

Answer

The column parameter c equals 0.8 for sawn lumber, 0.85 for round timber poles, and 0.9 for structural glued laminated timber and structural composite lumber. Selecting the wrong c for the member type meaningfully changes the resulting C_p value.

#228 · Wood Structures

Scenario: a wood post is unbraced 16 feet about its strong axis but braced at 8 foot intervals about its weak axis by a girt. Which axis governs slenderness?

Answer

Bracing only helps the axis it actually restrains, so the girt reduces the weak axis unbraced length but does nothing for the strong axis, which still runs the full 16 feet unbraced. Even though weak axis is usually more slender for an unbraced member, here the strong axis ends up with the larger L_e over d and governs.

#229 · Wood Structures

For a 5.5 inch square post unbraced 16 feet strong axis and 8 feet weak axis, roughly what are the two L_e over d ratios?

Answer

Strong axis: 192 inches divided by 5.5 inches gives about 35. Weak axis: 96 inches divided by 5.5 inches gives about 17. The strong axis, at roughly double the weak axis slenderness ratio here, is the one carried into the F_cE and C_p calculation.

#230 · Wood Structures

Common exam trap regarding column bracing shown in a floor plan exhibit: what should you not assume?

Answer

Do not assume that bracing shown, such as a girt line or a mid height diagonal brace, helps both axes of the column. Bracing only restrains the axis it is physically oriented to resist, so the other axis may still be governed by its full unbraced story height.

#231 · Wood Structures

Which NDS chapters govern wood connections such as bolts, lag screws, wood screws, nails, and split ring or shear plate connectors?

Answer

NDS Chapter 11 gives general mechanical connection provisions, Chapter 12 covers dowel-type fasteners (bolts, lag screws, wood screws, nails), and Chapter 13 covers split ring and shear plate connectors.

#232 · Wood Structures

What is the governing design method in NDS 12.3 for bolts and other dowel type fasteners?

Answer

NDS 12.3 uses the yield limit equation approach, which models several possible failure modes: wood bearing crushing in the main member, wood bearing crushing in the side member, and combinations of bolt bending with one or two plastic hinges forming.

#233 · Wood Structures

Which yield mode governs the final connection capacity Z , and why?

Answer

The lowest resulting Z value among all the yield mode equations governs. That is the failure mode that actually occurs first in the real connection, so the connection can never carry more than the weakest of the possible failure paths.

#234 · Wood Structures

Do you need to run all the NDS 12.3 yield limit equations by hand for a common bolted connection on exam day?

Answer

Not necessarily. The reference handbook includes pre-computed single fastener design value tables, like NDS Table 12A, organized by species group, fastener diameter, and member thickness. For common configurations, using this table is faster and a legitimate lookup-first strategy.

#235 · Wood Structures

After pulling a single fastener value from NDS Table 12A, which adjustment factors still need to be applied conceptually?

Answer

CD (load duration), C_g (the group action factor), and C_{Δ} (the geometry factor for spacing and end distance) still need to be applied afterward. These are not baked into the base table value.

#236 · Wood Structures

What does the group action factor C_g account for?

Answer

C_g accounts for the fact that fasteners arranged in a row do not share the applied load perfectly evenly among themselves. The end fasteners in a row tend to pick up more load than fasteners in the middle, so C_g reduces the effective capacity per fastener as row length or fastener count increases.

#237 · Wood Structures

What does the geometry factor C-delta penalize?

Answer

C-delta penalizes a connection when actual fastener spacing or end distance falls short of the tabulated full value baseline requirements. Tighter than ideal spacing or insufficient end distance reduces the usable fastener capacity below the tabulated single fastener value.

#238 · Wood Structures

Trap: which two adjustment factors do fasteners never receive, even though they apply to bending members?

Answer

Fasteners never get C_r , the repetitive member factor, or C_F , the size factor, since both of those belong strictly to bending members. Mixing these up is a classic wrong answer choice on a select all that apply connection item.

#239 · Wood Structures

How do lag screws behave under lateral loading compared to bolts?

Answer

Lag screws behave like bolts for lateral loading, meaning load applied perpendicular to the screw shank, and use the same NDS 12.3 yield limit framework as bolts for that direction of loading.

#240 · Wood Structures

What additional capacity do lag screws have beyond lateral capacity, and how is it different?

Answer

Lag screws also have a withdrawal capacity for loads applied parallel to the screw shank. This uses a completely different equation based on the screw's threaded engagement length into the wood, not the yield limit theory used for lateral loading.

#241 · Wood Structures

A deck ledger connects deck joists to a band joist, transferring vertical shear. Is this lag screw connection loaded laterally or in withdrawal?

Answer

This connection is loaded laterally, since the vertical shear transfer is perpendicular to the fastener shank, not axial pull-out along the shank. Confusing this with a withdrawal loaded condition is a real and recurring exam trap.

#242 · Wood Structures

How are nailed connections typically handled compared to bolts and lag screws?

Answer

Nails are governed by the same NDS 12.3 yield limit theory scaled down for smaller diameters, but they are typically tabulated directly by nail size and species group rather than run through the full yield mode equation set by hand.

#243 · Wood Structures

Where do nailed connections most commonly appear in an SE exam problem, rather than as a standalone fastener check?

Answer

Nailed connections almost always show up embedded inside a shear wall or diaphragm unit shear value, consumed through the SDPWS tables, rather than being calculated fastener by fastener as an isolated connection problem.

#244 · Wood Structures

Which document governs lateral wood systems such as shear walls and diaphragms?

Answer

SDPWS 2015, the Special Design Provisions for Wind and Seismic, governs lateral wood systems built from wood structural panels, including both shear walls and diaphragms.

#245 · Wood Structures

What does SDPWS Table 4.3A give you, and what is already built into those values?

Answer

Table 4.3A gives the ASD unit shear capacity for a given sheathing thickness, panel grade, nail size, nail spacing, and minimum framing width. The strength adjustment and safety factor are already incorporated into the tabulated ASD values, so this is a direct lookup, not something you derive from a nail shear equation.

#246 · Wood Structures

What aspect ratio limit applies to blocked wood structural panel shear walls under SDPWS 4.3.4?

Answer

Blocked wood structural panel shear walls are limited to a height to width ratio of 2 to 1 to use the tabulated unit shear capacity without reduction. Beyond that ratio, the wall's usable capacity gets penalized.

#247 • Wood Structures

Trap: a shear wall segment passes the unit shear capacity check from Table 4.3A. Is it automatically adequate?

Answer

Not necessarily. The aspect ratio limit in SDPWS 4.3.4 is a separate geometry check layered on top of the tabulated capacity. A wall that is unusually tall and narrow relative to its length can still fail this check even though its unit shear capacity looks sufficient.

#248 • Wood Structures

How does a diaphragm's structural action differ from a shear wall's, even though both use the same nailing tables?

Answer

A diaphragm acts as a horizontal beam spanning between shear walls, rather than as a vertical element. It develops shear along its length resisted by sheathing nailing, plus a bending moment resolved into chord forces at its boundary.

#249 · Wood Structures

How is diaphragm chord force calculated, and what is the analogous element in a steel or concrete beam?

Answer

Chord force equals the diaphragm moment divided by the diaphragm depth, meaning the dimension perpendicular to the span. This is directly analogous to the flange force in a steel or concrete beam, since the chord resists the tension or compression couple from bending.

#250 · Wood Structures

What physical members typically serve as diaphragm chords, and what must their splices be checked for?

Answer

Chords are usually existing framing members, like a double top plate or a ledger. Their splices require a net section check where fastener holes remove wood cross section, plus a check that the fastener group at the splice can transfer the calculated chord force.

#251 · Wood Structures

What does the glulam combination symbol 24F-1.8E tell you?

Answer

The 24F portion tells you the reference bending value, 2400 psi. The 1.8E portion tells you the reference modulus of elasticity, 1.8 million psi, for that specific glulam combination.

#252 · Wood Structures

What size related factor does glulam use instead of CF, and why is a different factor needed?

Answer

Glulam uses the volume factor CV instead of CF, because glulam manufacturing and material variability behave differently from sawn lumber. CV reflects statistical defect probability across a manufactured, layered product rather than a single sawn piece.

#253 · Wood Structures

How does CV change as glulam beam length, depth, or width increase?

Answer

CV decreases as any of those dimensions increases. This reflects a higher statistical chance of encountering a strength limiting defect somewhere along a larger piece of material.

#254 · Wood Structures

What is the critical rule from NDS 5.3.6 regarding CV and CL for a glulam beam?

Answer

CV and CL are never multiplied together. NDS 5.3.6 requires computing both factors and using only the smaller of the two in the design calculation, not their product.

#255 · Wood Structures

For a glulam beam with a continuously braced compression edge, which factor typically governs, CV or CL?

Answer

CL equals 1.0 for a continuously braced compression edge, so CV, which is normally well below 1.0 for any beam of real size, governs by default in that case since it is the smaller of the two.

#256 · Wood Structures

How are design values for structural composite lumber products like LVL and PSL typically provided on the exam?

Answer

LVL, PSL, and similar structural composite lumber products use manufacturer published or code recognized design values. These are typically supplied directly in an exhibit or table excerpt rather than derived from NDS Supplement tables like sawn lumber or standard glulam combinations.

#257 · Wood Structures

List the seven common exam traps from this chapter's closing summary, in brief.

Answer

CD is set by shortest duration not largest load, CM and Ct do not apply automatically to interior framing, column slenderness is governed by the larger L_e over d axis, fasteners never get Cr or CF, CV and CL are never multiplied, always check lateral versus withdrawal loading on a fastener, and shear wall aspect ratio is a separate check from unit shear capacity.

#258 · Wood Structures

What is the core skill this exam is actually testing regarding adjustment factors, according to the chapter?

Answer

The exam tests whether you can correctly interpret what a scenario is physically describing and then find the right equation or table with that interpretation already locked in. It does not test memorization of tabulated design values, since those are provided in the reference handbook.

#259 · Wood Structures

Why does CD equal 1.6 for a wind or seismic load combination rather than a larger value?

Answer

CD of 1.6 reflects the very short cumulative duration that wind or seismic loading is expected to act on a structure over its life. Wood fibers can sustain a higher stress for a very brief duration than for a sustained one, so shorter expected duration produces a larger CD multiplier on the reference value.

#260 · Wood Structures

Scenario: an exhibit shows a floor system of five parallel 2x10 joists at 16 inches on center sheathed with plywood. Does Cr apply to Fb?

Answer

Yes. This meets the Cr requirements: three or more parallel members, spacing at or under 24 inches on center, and connected by a load distributing element, the plywood sheathing. Cr equal to 1.15 applies to Fb only, not to Ft, Fv, or Fc.

#261 · Wood Structures

What does it mean physically that F_c perpendicular has no C_p column stability factor applied to it?

Answer

C_p addresses buckling of a column under axial compression parallel to grain, a stability failure mode. Bearing perpendicular to grain is a localized crushing check at a support, not a buckling phenomenon, so C_p is never part of an F_c perpendicular bearing check.

#262 · Wood Structures

If a beam's design shear is checked at a distance d from the support face under NDS 3.4.3, what happens to loads within that distance d ?

Answer

Concentrated loads or the portion of uniform load located within distance d of the support face are excluded from the shear computed at that reduced location, since they are assumed to be carried more directly into the support without fully loading the beam in shear there.

#263 · Wood Structures

Why should a candidate not automatically assume a size factor of 1.0 for a glulam member the way they might handle CF for sawn lumber?

Answer

Glulam does not use CF at all. It uses CV, the volume factor, which is essentially never simply 1.0 for a real sized beam since it decreases with length, depth, and width. Treating glulam sizing like sawn lumber sizing is a category error.

#264 · Wood Structures

What role does the section modulus S play in the bending stress check, and what governs which S value to use?

Answer

S is the section modulus of the beam's actual cross section, used in f_b equals M over S . For a rectangular sawn lumber section this comes from standard dimensions, while for glulam or engineered products it comes from the actual manufactured depth and width given in the exhibit or reference table.

#265 · Wood Structures

Trap: a candidate applies CM to a glulam beam supporting an outdoor covered walkway roof with no direct rain or ground contact. Should CM apply?

Answer

Only if the actual in service moisture content is expected to exceed 19 percent sustained. A covered, protected exterior location may or may not qualify depending on the real exposure described, so the decision must be based on the physical scenario given, not a blanket rule for anything labeled exterior.

#266 · Wood Structures

How does the group action factor C_g generally trend as more fasteners are added to a single row?

Answer

C_g generally decreases as the number of fasteners in a row increases, because load sharing becomes less uniform across more fasteners in series. This means simply adding more bolts to a row does not increase per fastener efficiency proportionally.

#267 · Wood Structures

What is the practical difference between checking bolt capacity via the NDS 12.3 yield equations versus NDS Table 12A?

Answer

The yield equations in 12.3 derive Z from first principles across multiple possible failure modes and are more general and time consuming. Table 12A gives pre-computed single fastener values for common species, diameter, and thickness combinations, which is faster for typical configurations already covered by the table.

#268 · Wood Structures

Why can a column with an L_e over d ratio of 55 never be brought into compliance just by increasing F_c star through species selection?

Answer

NDS 3.7.1.4 caps L_e over d at 50 for sawn lumber columns as an absolute geometric limit. A ratio of 55 fails regardless of material strength, since the limit addresses buckling geometry, not material capacity, so a stronger species cannot fix a slenderness violation.

#269 · Wood Structures

What is the conceptual difference between CL for a sawn lumber or glulam beam and Cp for a column?

Answer

CL addresses lateral torsional buckling of a bending member's unbraced compression edge, a flexural instability mode. Cp addresses axial buckling of a column, a compression member instability mode. Both are stability factors but apply to physically different failure mechanisms and different member types.

#270 · Wood Structures

In the column parameter c values, which member type uses 0.9, and why might that be different from sawn lumber's 0.8?

Answer

Structural glued laminated timber and structural composite lumber use c equal to 0.9, reflecting their more consistent, engineered material properties compared to natural sawn lumber's c of 0.8. Round timber poles fall in between at 0.85.

#271 · Wood Structures

What is the difference between checking a shear wall's unit shear capacity and checking its overturning or hold down demand?

Answer

Unit shear capacity from SDPWS Table 4.3A addresses the sheathing nailing's ability to resist lateral shear along the wall length. Overturning and hold down demand address the tension and compression couple at the wall ends from the applied lateral force, a separate check this chapter's aspect ratio discussion sits alongside but does not replace.

#272 · Wood Structures

Why is a diaphragm chord splice net section check necessary even if the gross chord member size looks adequate for the calculated force?

Answer

Fastener holes at the splice remove wood cross section, reducing the effective area available to resist the chord tension force at exactly the location where the connection also has to transfer that same force. The gross section away from the splice is not the governing check location.

#273 · Wood Structures

Why does the exam favor scenario based interpretation over factor memorization when adjustment factors are involved?

Answer

Every reference design value and factor definition is available in the electronic handbook, but recognizing which physical conditions in a given scenario trigger or exclude a factor is not something a table lookup can tell you. That interpretive judgment, not recall, is what the exam is actually built to test.

#274 · Wood Structures

A lag screw connection in a hanging sign bracket pulls the screw straight out of the wood along its axis. Is this lateral or withdrawal loading?

Answer

This is withdrawal loading, since the load acts parallel to the screw shank, pulling it out along its threaded engagement length. This requires the withdrawal capacity equation, not the lateral yield limit equations used for perpendicular loading.

#275 • Wood Structures

Summarize the overall study strategy this chapter recommends for the Wood Structures depth topic.

Answer

Spend less time memorizing tabulated design values since the reference handbook supplies them during the exam. Spend more time practicing recognition of which adjustment factors, failure modes, and loading directions a given physical scenario actually calls for, since that judgment is the real tested skill.

#276 • Masonry Structures

What two documents govern masonry design and construction under the codes used on this exam?

Answer

TMS 402 covers design provisions and TMS 602 covers specification, material, and construction requirements. They work as a pair, with TMS 402 telling you how to calculate strength and TMS 602 telling you how the materials must be supplied and built. The exam's electronic reference handbook gives you searchable access to both during the test.

#277 · Masonry Structures

Is f'_m a property of the masonry unit alone or of the assembly?

Answer

f'_m , the specified compressive strength of masonry, is a property of the whole assembly, not just the block or brick unit. It depends on unit strength, mortar type, and grout, and can be verified either by testing or by the unit-strength method in TMS 602. Almost every exam problem hands you f'_m directly, so you rarely need to derive it.

#278 · Masonry Structures

List the four mortar types recognized by TMS 602 in decreasing strength order.

Answer

The four mortar types are M, S, N, and O, from strongest to weakest. Type M is the strongest and most durable, while Type O is the weakest and generally limited to non-structural or interior applications. Mortar type is one of the inputs that determines the resulting f'_m of the assembly.

#279 · Masonry Structures

Why does grout get added to hollow masonry units?

Answer

Grout fills the cells of hollow units to bond reinforcing steel to the masonry and to increase the effective cross sectional area that resists load. Without grout, a bar placed in an empty cell would have no way to transfer force into the surrounding masonry. Grouting is what turns a hollow unit wall into a composite reinforced section.

#280 · Masonry Structures

Define fully grouted versus partially grouted masonry.

Answer

Fully grouted masonry has every cell filled with grout, giving the section its full gross cross sectional area. Partially grouted masonry only fills the cells that contain reinforcement, commonly spaced at 16, 24, 32, or 48 inches on center, leaving the remaining cells hollow. This distinction changes almost every downstream strength calculation.

#281 · Masonry Structures

In a partially grouted wall, what area is used for out-of-plane flexural or shear calculations, and why is this a common exam trap?

Answer

Only the grouted cell area plus the face shells count, similar in concept to a T-beam flange rather than a solid rectangle. Candidates frequently default to the full gross wall thickness out of habit, which overstates capacity. Recognizing the grouting pattern in the problem statement before picking an effective width is the key first step.

#282 · Masonry Structures

An 8-inch nominal CMU is actually how thick, and why does the difference matter?

Answer

An 8-inch nominal CMU is actually 7.625 inches thick, since nominal masonry dimensions run $\frac{3}{8}$ inch larger than actual to account for mortar joints. TMS 402 strength equations all use actual dimensions, so a problem giving a nominal size must be converted before any calculation. Skipping this conversion shifts every downstream number and can flip which multiple-correct answers are true.

#283 · Masonry Structures

A 16-inch nominal CMU length converts to what actual dimension?

Answer

A 16-inch nominal length is actually 15.625 inches, following the same 3/8-inch mortar-joint reduction rule used for thickness. Any spacing or length value stated as a nominal block dimension needs this same conversion before it goes into a TMS 402 formula. This applies to length just as much as it applies to wall thickness.

#284 · Masonry Structures

What grade of reinforcing steel is almost always used in masonry design, and what is f_y ?

Answer

Masonry reinforcement is almost always Grade 60 deformed bar, giving $f_y = 60$ ksi. The bars sit inside grouted cells or bond beams to develop tension capacity paired with the masonry compression zone. This is the same f_y value used throughout the strength design flexural and axial equations in TMS 402 Chapter 9.

#285 · Masonry Structures

Can a masonry wall pass its strength calculation and still fail the exam problem?

Answer

Yes. TMS 402 sets minimum and maximum reinforcement ratios and bar spacing rules on top of the strength check, and Chapter 7 adds prescriptive detailing requirements in higher seismic design categories. A wall that satisfies bending strength alone can still violate a seismic detailing minimum, so both checks are required, not just the stress calculation.

#286 · Masonry Structures

Which TMS 402 chapters cover Allowable Stress Design and Strength Design, and are the two methods ever mixed on one project?

Answer

Allowable Stress Design is Chapter 8 and Strength Design is Chapter 9. They are two complete, independent design paths and are not mixed together within a single project. A designer picks one philosophy and uses it consistently for that structure.

#287 · Masonry Structures

How does ASD compare demand and capacity in masonry design?

Answer

ASD compares service-level stresses, computed from unfactored loads taken straight from ASCE 7-16, against allowable stresses that already have a built-in safety factor. There are no separate load factors or phi factors applied on top of that comparison. The safety margin lives inside the allowable stress value itself.

#288 · Masonry Structures

How does Strength Design compare demand and capacity in masonry design?

Answer

Strength Design compares a factored demand, built from ASCE 7-16 load combinations with load factors, against a nominal strength multiplied by a phi factor. This mirrors the same basic philosophy used in ACI 318 concrete design. The safety margin is split between the load-side factors and the phi capacity reduction factor.

#289 · Masonry Structures

In the first ten seconds of a masonry problem, how can you tell whether it wants ASD or Strength Design?

Answer

If the problem gives unfactored dead, live, and wind loads with no phi factors mentioned, that signals ASD. If it gives already-factored P_u and M_u values with phi factors referenced, that signals Strength Design. Reading this signal early saves time because it tells you which handbook chapter to open first.

#290 · Masonry Structures

What masonry compressive stress value is used in the Strength Design rectangular stress block, and how does it differ from concrete?

Answer

Masonry Strength Design uses $0.80 f'_m$ distributed over a depth a , compared to concrete's $0.85 f'_c$ factor in ACI 318. The two materials use different coefficients because masonry behaves differently near compression failure than concrete does. Carrying the concrete 0.85 factor into a masonry problem out of habit is a classic wrong-answer trap.

#291 · Masonry Structures

Where does the 0.80 f'_m stress block coefficient show up most often on the exam?

Answer

It appears in nearly every strength design flexural equation in TMS 402 Chapter 9, anywhere a rectangular compression block depth a is computed. Since it looks so similar to concrete's 0.85 f'_c , mixing the two up is one of the most common ways to get a fast, plausible-looking wrong answer. Re-checking this number is worth the extra few seconds any time a concrete problem appears nearby in the same session.

#292 · Masonry Structures

Write the formula for the compression block depth a in a reinforced masonry flexural section under Strength Design.

Answer

a equals A_s times f_y , divided by 0.80 f'_m times b , where A_s is the tension steel area, f_y is the steel yield strength, f'_m is the masonry compressive strength, and b is the design strip width. For a wall analyzed per foot of length, b is often taken as 12 inches. This a value then feeds directly into the nominal moment strength equation.

#293 · Masonry Structures

Write the nominal flexural strength equation M_n for a reinforced masonry section, and define d .

Answer

M_n equals A_s times f_y times the quantity d minus a divided by 2, where d is the distance from the extreme compression face to the centroid of the tension steel. For a solid wall with steel centered in the cross section, d is often close to half the actual wall thickness. This is the same internal force couple logic used in reinforced concrete flexural design.

#294 · Masonry Structures

What ϕ factor applies to a tension-controlled masonry flexural section, and what condition must be checked?

Answer

TMS 402 Chapter 9 assigns $\phi = 0.9$ for flexure in most practical tension-controlled wall cases. The check required is that ϕM_n meets or exceeds M_u , the factored moment demand. This is the final capacity comparison after A_s , a , and M_n have all been computed.

#295 · Masonry Structures

A masonry lintel is 7.625 inches wide, $d = 13.5$ inches, $f'm = 2,500$ psi, reinforced with two number 6 bars ($A_s = 0.88$ in squared), Grade 60 steel. Find a .

Answer

a equals A_s times f_y divided by $0.80 f'm b$, which is 0.88 times 60 divided by 0.80 times 2.5 times 7.625 . That is 52.8 divided by 15.25 , giving a equal to about 3.46 inches. This a value is then used directly to find the nominal moment strength.

#296 · Masonry Structures

Continuing the same lintel ($A_s = 0.88$ in squared, $f_y = 60$ ksi, $d = 13.5$ in, $a = 3.46$ in), find M_n and ϕM_n .

Answer

M_n equals 0.88 times 60 times the quantity 13.5 minus 1.73 , which is 52.8 times 11.77 , giving about 621.4 kip-inches, or 51.8 kip-feet. Applying $\phi = 0.9$ gives ϕM_n of about 46.6 kip-feet. Every step in this chain is a fast hand calculation once the setup with actual dimensions is correct.

#297 · Masonry Structures

What is the eccentricity e in a masonry axial-plus-bending problem under Strength Design, and what does it represent?

Answer

Eccentricity e equals M_u divided by P_u , the factored moment demand divided by the factored axial demand. It represents how far outside the geometric centerline the resultant axial force effectively acts. A large e means the load behaves more like pure bending, while a small e means the section behaves more like pure axial compression.

#298 · Masonry Structures

Define the kern of a rectangular masonry section and give its width.

Answer

The kern is the zone within which an axial load produces no tension anywhere in the section, and it extends t divided by 6 on either side of the section centerline, where t is the actual section depth in the direction of bending. As long as the eccentricity e stays inside this zone, the wall stays in compression across its full cross section under service loads. This is the same kern concept used in unreinforced or lightly loaded masonry and concrete sections generally.

#299 · Masonry Structures

What happens if the eccentricity e falls outside the kern of a masonry wall or column section?

Answer

The section is expected to crack in tension on one face under service loads, so a simple no-tension assumption is no longer valid. A full cracked-section analysis or a P-M interaction check is required instead. Recognizing whether e is inside or outside $t/6$ in the first thirty seconds of a column or pilaster problem determines which calculation path the rest of the problem needs.

#300 · Masonry Structures

Why is checking e against $t/6$ described as a fast first screen on axial-plus-bending masonry problems?

Answer

It tells you immediately whether the simpler no-tension design path applies or whether a more involved cracked-section or interaction check is required. Doing this comparison early avoids wasting time setting up the wrong calculation method. It is a quick arithmetic check that reshapes the entire remaining approach to the problem.

#301 · Masonry Structures

Write the nominal shear strength equation for a masonry shear wall under TMS 402 Chapter 9.

Answer

V_n equals V_{nm} plus V_{ns} , the sum of a masonry contribution and a reinforcement contribution. This mirrors the same additive shear strength logic used in reinforced concrete design. Each term is computed separately before being added together and compared against the factored shear demand.

#302 · Masonry Structures

What ratio controls the masonry shear coefficient in TMS 402, and what does it physically represent?

Answer

The ratio μ divided by V_u times d_v , where d_v is the shear depth, controls the coefficient applied to the masonry shear contribution. A low ratio means the wall behaves like a deep, squat shear element with high shear capacity, while a high ratio (capped at a maximum of 1.0) means it behaves like a slender flexural element with lower shear capacity. The code coefficient is bounded between 2.25 and 4.0 to reflect this range of behavior.

#303 · Masonry Structures

How is the shear depth d_v typically taken for a rectangular masonry shear wall?

Answer

d_v is typically taken as 0.8 times the wall length L_w . This value is used directly in the μ_u divided by $V_u d_v$ ratio that sets the masonry shear coefficient. It represents an effective lever arm rather than the full wall length.

#304 · Masonry Structures

An 8-inch (7.625 in actual) fully grouted CMU shear wall is 20 feet long. Find the shear area A_{nv} .

Answer

A_{nv} equals the actual thickness times the wall length, so 7.625 times (20 times 12 inches) equals 7.625 times 240, which is 1,830 square inches. This uses the actual 7.625 inch thickness, not the nominal 8 inch value. A_{nv} is the area multiplied by the masonry shear coefficient and the square root of f'_m in the V_{nm} equation.

#305 · Masonry Structures

For the same 20-foot-long, 12-foot-tall cantilevered shear wall with $V_u = 45$ kips, find M_u and the ratio $M_u / (V_u d_v)$.

Answer

M_u equals V_u times h , so 45 times 12 equals 540 kip-feet at the base of the cantilever. With $d_v = 0.8$ times 20 = 16 feet, the ratio M_u divided by $(V_u d_v)$ equals 540 divided by (45 times 16), which is 540 divided by 720, or 0.75. That value falls within the code's normal 1.0 to 4.0 coefficient range check before being applied.

#306 · Masonry Structures

Using $M_u / (V_u d_v) = 0.75$, find the masonry shear coefficient and V_{nm} for the 1,830 square inch, $f'_m = 2,000$ psi shear wall.

Answer

The coefficient is 4.0 minus 1.75 times 0.75, which equals 2.6875. Then V_{nm} equals 2.6875 times 1,830 times the square root of 2,000 psi, which is 2.6875 times 1,830 times 44.72, giving about 219,900 pounds or 219.9 kips.

#307 · Masonry Structures

What is the design shear strength ϕV_m for the wall above, and how does it compare to the 45 kip demand?

Answer

Applying the shear ϕ factor of 0.8 to V_m of 219.9 kips gives a design masonry shear strength of about 176 kips. That is far above the 45 kip factored demand, showing the wall is shear-adequate from the masonry contribution alone, before any reinforcement contribution is even added. This illustrates how quickly a fully grouted, generously sized shear wall can pass shear even without counting steel.

#308 · Masonry Structures

What triggers the need for boundary elements in a masonry shear wall?

Answer

Boundary elements come into play for taller, more slender shear walls or walls in higher seismic design categories. TMS 402 Chapter 7 can require concentrated vertical reinforcement and confinement at the wall ends in these cases. Their purpose is to prevent compression toe crushing under combined axial load and overturning moment.

#309 · Masonry Structures

What clue in a shear wall problem signals that special boundary element provisions should be checked?

Answer

A tall, narrow shear wall combined with a high overturning moment relative to its length is the signal to check special boundary element requirements. Passing the basic shear equation alone is not sufficient in that scenario. The problem is testing whether the candidate recognizes this trigger, not just whether they can run the shear formula.

#310 · Masonry Structures

How are masonry lintels designed structurally, compared to a wall in bending?

Answer

Lintels are designed as reinforced flexural members using the exact same rectangular stress block logic as a wall in bending. The only real difference is that a lintel has a discrete beam-shaped cross section rather than being analyzed as a per-foot wall strip. The 0.80 f_m stress block and M_n equation apply identically.

#311 · Masonry Structures

Why does bearing length matter at the ends of a masonry lintel?

Answer

TMS 402 sets a minimum bearing length, commonly 4 inches and more in high seismic areas, to make sure the reaction force does not crush the masonry locally at the support. Bearing length also affects how the effective span is defined for the flexural calculation. Both the minimum bearing check and the span definition need to be confirmed from the problem's exhibit.

#312 · Masonry Structures

How is the effective span length of a masonry lintel usually defined?

Answer

Effective span is usually taken as the clear span plus some fraction of the bearing length on each side, rather than clear span alone or full center-to-center support distance. Because this varies by problem setup, the exhibit needs to be read carefully to see which span definition it intends. Using the wrong span length shifts the moment demand calculation directly.

#313 · Masonry Structures

What does veneer anchor design in TMS 402 Chapter 12 use as its basic design concept?

Answer

Veneer anchor design works on a tributary area concept. Each anchor has a tested or specified capacity, wind pressure creates a demand per square foot, and dividing capacity by demand gives the maximum tributary area one anchor can serve. This tributary area then determines the anchor spacing needed across the veneer wall.

#314 · Masonry Structures

What is the classic exam trap involving veneer anchor tributary area?

Answer

TMS 402 caps the tributary area at a fixed maximum, 2.67 square feet in the common case, no matter how strong the anchor or how low the wind pressure calculates out to be. Candidates often calculate a generous tributary area from wind pressure alone and forget this flat code maximum, ending up with too few anchors specified. Whichever value is smaller, the calculated demand-based tributary area or the code maximum, always governs.

#315 · Masonry Structures

What two kinds of forces do masonry-to-diaphragm connections need to transfer?

Answer

These connections transfer out-of-plane wind or seismic pressure, where the wall pushes into the diaphragm, and in-plane shear, where the diaphragm reaction pushes back into the wall. Both load paths need to be checked at the same connection, since they act in different directions. Missing one direction of transfer leaves an incomplete load path.

#316 · Masonry Structures

Why do flexible diaphragms increase the required wall anchorage force in higher seismic design categories?

Answer

ASCE 7-16 has specific provisions for wall anchorage to flexible diaphragms because flexible diaphragms amplify demand near their edges compared to a simple tributary calculation. In higher seismic design categories, this amplification requires anchorage forces above what a basic tributary calculation alone would give. A candidate using only tributary area for this connection in a high seismic design category would understate the actual demand.

#317 · Masonry Structures

How are out-of-plane loaded bearing or non-bearing walls typically modeled structurally?

Answer

They are typically modeled as simply supported, one-way spanning members between floor and roof levels. Wind or seismic pressure acts as a uniform load across that pin-pin span. This gives a standard maximum moment equation of M_u equals w_u times h squared divided by 8.

#318 · Masonry Structures

A wall spans $h = 20$ feet with a strength-level wind pressure of 30 psf. Find M_u per foot of width.

Answer

M_u equals w_u times h squared divided by 8, so 30 times 400 divided by 8 equals 1,500 pound-feet per foot of width, or 1.5 kip-feet per foot. That moment is then checked the same way as any other flexural section, using actual thickness and effective depth. This is the standard one-way out-of-plane wall moment setup.

#319 · Masonry Structures

What ratio is used as a fast first screen for out-of-plane wall slenderness, and what value signals a concern?

Answer

The ratio h over t , unsupported height divided by actual thickness, is the fast first screen. A wall with h over t past roughly 30 is a signal that second-order, or P-delta, effects may need to be considered rather than ignored. Computing this ratio first tells you whether the problem is testing basic flexure or testing recognition of a slenderness trigger.

#320 · Masonry Structures

Define P-delta effect in the context of an out-of-plane masonry wall.

Answer

P-delta effect is the extra moment created when axial load acts through the lateral deflection of a slender wall, adding on top of the applied lateral load moment. It becomes significant as h over t increases past roughly 30. TMS 402 gives explicit methods to account for it rather than leaving it to judgment.

#321 · Masonry Structures

Name the two approaches TMS 402 gives for handling slender out-of-plane wall design.

Answer

TMS 402 gives a moment magnifier approach and, for slender or lightly loaded walls, an alternative deflection-based design method. Both exist to capture second-order P-delta effects that a simple $\mu = wh^2/8$ check would miss. Choosing between the two depends on the wall's loading and slenderness situation described in the problem.

#322 · Masonry Structures

Why should axial load at wall mid-height always include self-weight above that section?

Answer

The axial demand at mid-height must include the wall's own self-weight tributary above that point, not just the load coming down from the roof or floor above. Skipping this addition understates axial demand at exactly the section where slenderness effects are largest. It is a small numeric error that is easy to make under time pressure.

#323 · Masonry Structures

A wall carries 2.4 kips per foot from the roof, has partially grouted self-weight of 48 psf, and 10 feet of wall above the mid-height section checked. Find total axial demand at that section.

Answer

Self-weight adds 48 times 10 divided by 1,000, which is 0.48 kips per foot of additional dead load. Adding that to the 2.4 kips per foot from the roof gives a total of 2.88 kips per foot at the mid-height section. This total, not just the roof load alone, is what should be checked against axial and interaction capacity.

#324 · Masonry Structures

What is the single most common masonry mistake candidates make on this exam?

Answer

The most common mistake is using nominal dimensions instead of actual dimensions somewhere in a calculation, most often wall thickness or bar cover. Since nominal sizes run 4 inches larger than actual, this error shifts every downstream number in the wrong direction. Converting to actual dimensions should be the first step in any masonry calculation.

#325 · Masonry Structures

What is the second most common masonry mistake on this exam?

Answer

The second most common mistake is forgetting that partially grouted walls only get credit for grouted cell area and effective flange width, not the gross cross section, when computing shear area or flexural stress block width. This applies whether the calculation is for out-of-plane flexure or in-plane shear. Confirming the grouting pattern before setting up the area term avoids this error.

#326 · Masonry Structures

What is the third most common masonry mistake, and when is a candidate most at risk of making it?

Answer

The third most common mistake is mixing up the 0.80 f'_m masonry stress block coefficient with the 0.85 f'_c coefficient used in concrete design. Candidates are most at risk right after finishing a concrete problem in the same exam session, when the concrete factor is still fresh in memory. Actively re-checking this number before using it in a masonry problem prevents the error.

#327 · Masonry Structures

Describe the fourth common masonry exam trap involving code caps and minimums.

Answer

The fourth trap is applying a calculated result without checking the code's built-in cap or minimum on that result, such as veneer anchor tributary area, minimum reinforcement ratios in seismic design categories, or maximum reinforcement spacing limits. TMS 402 frequently gives a formula and then a separate limiting clause in the same section. A candidate can compute the formula correctly and still get a multiple-correct question wrong by never reading the limiting clause.

#328 · Masonry Structures

What is the recommended time-management workflow for a scenario-based masonry problem?

Answer

Extract every given number from the scenario text and exhibit into a short list before opening the handbook, decide up front whether the scenario is ASD or Strength Design, then look up only the one or two equations actually needed rather than re-reading whole code sections. Masonry equations are long but mechanical once inputs are organized. Most of the time saved comes from setup discipline, not calculator speed.

#329 · Masonry Structures

Why can you not memorize TMS 402 shear and flexure equations word for word and expect that to be enough on this exam?

Answer

You are not expected to have Chapter 9 of TMS 402 memorized, since the electronic reference handbook gives searchable access to the actual text during the exam. What you are expected to know is which equation applies to which situation, what every variable means physically, and how to set the numbers up fast. The exam rewards recognition and setup speed, not rote memorization of the formula text.

#330 · Masonry Structures

A problem states a wall is partially grouted with reinforcement at 32 inches on center. What does this tell you about which cells are grouted?

Answer

Only the cells containing the vertical reinforcement bars are filled with grout, spaced at 32 inches on center, while the cells between them remain hollow. This means the effective area for strength calculations is limited to those grouted cells plus the face shells, not the full wall length. The 32 inch spacing itself also needs to be checked against any applicable maximum reinforcement spacing limit.

#331 · Masonry Structures

If a masonry problem gives you M_u and P_u directly with a stated ϕ factor, what design method are you in?

Answer

You are in Strength Design, since factored moment and axial demand paired with ϕ factors is the signature of TMS 402 Chapter 9. This tells you to reach for the 0.80 f'_m stress block equations rather than ASD allowable stress tables. Recognizing this signal quickly avoids wasted time in the wrong handbook section.

#332 · Masonry Structures

What does a low $M_u/(V_u d_v)$ ratio near 0 mean physically for a masonry shear wall?

Answer

A low ratio near 0 means the wall behaves like a deep, squat shear element with high shear capacity relative to its length. This corresponds to the code's maximum shear coefficient bound of 4.0.

#333 · Masonry Structures

What does a high $M_u/(V_u d_v)$ ratio near its code-capped maximum of 1.0 mean physically for a masonry shear wall?

Answer

A high ratio approaching its code-capped maximum of 1.0 means the wall behaves like a slender flexural element with lower shear capacity relative to its length. This corresponds to the code's minimum shear coefficient bound of 2.25.

#334 · Masonry Structures

Why is d_v taken as $0.8 L_w$ rather than the full wall length L_w in the shear equations?

Answer

d_v represents an effective shear depth or lever arm rather than the geometric wall length itself, similar in concept to using an effective depth rather than gross depth in flexural design. Using $0.8 L_w$ accounts for the fact that the full length is not fully effective in resisting shear. This value feeds into both the $M_u/(V_u d_v)$ ratio and the V_{nm} equation.

#335 · Masonry Structures

In the V_{nm} equation, what three quantities are multiplied together to get the masonry shear contribution?

Answer

V_{nm} is the product of the masonry shear coefficient (from the μ/ν dv ratio, bounded 2.25 to 4.0), the shear area A_{nv} , and the square root of f'_m . All three terms must be computed correctly and in consistent units before multiplying. Missing the coefficient bound check or using nominal instead of actual dimensions in A_{nv} are both common setup errors here.

#336 · Masonry Structures

What phi factor is typically applied to masonry shear strength, and where in the calculation does it get applied?

Answer

A shear phi factor of 0.8 is applied to the nominal shear strength V_n to get the design shear strength ϕV_n . This comes after V_{nm} and V_{ns} have been summed, as the final step before comparing against the factored shear demand V_u . In the worked shear wall example, ϕV_{nm} of about 176 kips comfortably exceeded the 45 kip demand.

#337 · Masonry Structures

Why does the exam expect you to check bar spacing and reinforcement ratio limits even after a flexural strength check passes?

Answer

TMS 402 sets minimum and maximum reinforcement ratios and spacing rules independently of the strength calculation, and Chapter 7 layers additional prescriptive detailing requirements on top in higher seismic design categories. A section can satisfy M_n versus M_u and still fail these separate detailing checks. Both the strength check and the detailing check are graded as part of a complete design.

#338 · Masonry Structures

What is the physical difference between a bearing wall and a non-bearing wall in out-of-plane design context?

Answer

A bearing wall carries vertical gravity load from floors or roof above in addition to out-of-plane lateral pressure, while a non-bearing wall primarily resists only the out-of-plane lateral pressure itself. Both are typically modeled as simply supported one-way spanning members between floor and roof levels for lateral load. The bearing wall case adds the axial load and eccentricity interaction on top of the pure flexural check.

#339 · Masonry Structures

How does the wall design d value relate to actual wall thickness for a solid wall with centered reinforcement?

Answer

For a solid wall with steel centered in the cross section, the effective depth d is often close to half the actual wall thickness, since the bar sits near the mid-plane of the wall. This is a useful quick estimate when a problem does not explicitly state d . Always confirm from the given cross section rather than assuming this relationship blindly if bar placement is off center.

#340 · Masonry Structures

Why is grout described as behaving like a T-beam flange concept in partially grouted wall analysis?

Answer

Only the grouted cells plus the adjacent face shells act as the effective resisting section, much like only the flange width of a T-beam contributes rather than the full slab width. The rest of the hollow, ungrouted wall contributes little to the strength calculation. This analogy helps candidates remember to limit the effective area rather than using the full gross wall thickness by habit.

#341 · Masonry Structures

A masonry column has e computed as slightly greater than $t/6$. What design path does this trigger?

Answer

Since e falls outside the kern, a simple no-tension assumption is no longer valid, and the section is expected to crack in tension on one face under service loads. This triggers the need for a cracked-section analysis or a full P-M interaction check rather than a basic axial-only calculation. Recognizing this early determines the rest of the solution path for the column problem.

#342 · Masonry Structures

What load combination source feeds both ASD and Strength Design in masonry, even though the two methods diverge afterward?

Answer

ASCE 7-16 provides the load combinations for both methods, unfactored service combinations for ASD and factored strength combinations for Strength Design. TMS 402 then applies its own allowable stress or phi-factor framework on top of whichever load set is used. The starting point for gathering loads is the same code either way, only the downstream combination and safety factor approach differ.

#343 · Masonry Structures

Why is reading the given exhibit important for determining a lintel's effective span, rather than always defaulting to clear span?

Answer

Effective span for a lintel is usually clear span plus some fraction of the bearing on each side, but the exact fraction and definition can vary by problem setup and exhibit. Defaulting to clear span alone, or to full center-to-center support distance, can misstate the moment demand. The problem's exhibit should be checked each time rather than relying on a fixed assumption.

#344 · Masonry Structures

In the shear wall worked example with $f'm = 2,000$ psi, why is the square root of $f'm$ taken in psi rather than in ksi?

Answer

TMS 402's shear strength equations are calibrated with $f'm$ in psi so the square root of $f'm$ produces a stress term in a consistent psi-based unit system alongside the shear area in square inches. Using ksi instead of psi under the square root would give a numerically wrong, much smaller result. Keeping units consistent through the square root step is a common place for unit errors to creep into shear calculations.

#345 · Masonry Structures

What is the practical exam-taking reason the reference handbook matters more in masonry than in almost any other Buildings Depth topic?

Answer

The equations for shear strength, flexural strength, and slenderness reduction all live inside TMS 402 and TMS 602 text, and candidates are not expected to memorize them, only to know which equation to look up and how to use it correctly. This makes masonry uniquely dependent on quick, accurate handbook navigation compared to topics where key equations are simple enough to memorize outright. Practicing the lookup and setup process ahead of time, not memorizing the formulas, is the more valuable exam preparation strategy here.

#346 · Masonry Structures

If a masonry wall problem gives you a nominal 6-inch block thickness, what actual thickness should you use in every equation?

Answer

A nominal 6-inch block converts to an actual thickness of 5.625 inches, following the same 3/8-inch reduction rule applied to all standard CMU nominal sizes. This actual value, not the nominal 6 inches, belongs in the shear area, stress block, and slenderness ratio calculations. Forgetting this conversion is the single most common masonry mistake on the exam.

#347 · Masonry Structures

Why can a candidate not simply assume h/t less than 30 automatically makes P-delta effects irrelevant?

Answer

The h/t past roughly 30 threshold is described as a fast first screen and signal to consider second-order effects, not a hard cutoff below which they can never matter. TMS 402 still provides the moment magnifier and deflection-based methods for cases that call for closer scrutiny even near that threshold. The safest habit is to always compute h/t first and use it to decide how carefully to examine slenderness, rather than treating 30 as an absolute pass/fail line.

#348 · Masonry Structures

What is the relationship between mortar type and f'_m when a problem uses the unit-strength method?

Answer

Under the unit-strength method, f'_m is determined from tables in TMS 602 that combine unit strength with the mortar type used, M, S, N, or O. Stronger mortar types combined with stronger units generally produce a higher resulting f'_m . This method is an alternative to direct assembly testing for establishing the design f'_m value.

#349 · Masonry Structures

In the reinforced masonry flexural equation, what happens to M_n if A_s is increased while everything else stays constant?

Answer

Increasing A_s increases a , since a equals $A_s f_y$ divided by $0.80 f'_m b$, which slightly reduces the internal lever arm (d minus $a/2$). However, the direct increase in A_s in the M_n equation more than offsets the small reduction in lever arm, so M_n increases overall with additional steel, up to the point where the section becomes over-reinforced or hits a maximum reinforcement ratio limit. This is the same general steel-versus-lever-arm tradeoff seen in reinforced concrete flexural design.

#350 · Masonry Structures

Summarize the correct order of operations for approaching an unfamiliar masonry Buildings Depth scenario under time pressure.

Answer

First extract every given number from the scenario and exhibit into a short list, converting any nominal dimensions to actual dimensions immediately. Second, decide whether the scenario is ASD or Strength Design based on whether loads and stresses or factored demands and ϕ factors are given. Third, look up only the specific equation needed in the handbook, apply it with actual dimensions and the correct $0.80 f'_m$ coefficient, and finally check any related code caps or minimums before finalizing the answer.

#351 · Foundations and Retaining Structures

What load level (ASD or LRFD) is used to check bearing pressure under a spread footing?

Answer

Bearing pressure is always checked with unfactored, service-level loads, not factored LRFD loads. The soil is compared against an allowable bearing pressure that is itself a service-level value. This is a classic trap because the same footing then switches to factored loads for the concrete shear and flexure design.

#352 · Foundations and Retaining Structures

Once a footing plan area is sized for bearing, what load level governs the rest of the ACI 318-14 design?

Answer

Everything after sizing the footing footprint uses factored loads and factored net soil pressure. The footing behaves like an upside down slab loaded by factored soil pressure pushing up against the column. Shear and flexure both use this factored pressure, not the service pressure used to set the footing area.

#353 · Foundations and Retaining Structures

Where is the critical section for two-way (punching) shear in a spread footing?

Answer

The critical section is located at a distance $d/2$ from the face of the column, where d is the effective depth of the footing. The critical perimeter, called b_o , wraps around all four sides of that offset rectangle. Getting b_o wrong is one of the single most common footing errors on this exam.

#354 · Foundations and Retaining Structures

Does one-way shear or two-way shear usually govern footing thickness?

Answer

Two-way (punching) shear almost always governs the thickness of a typical square footing. One-way shear rarely governs for square footings but can control for narrow strip or rectangular footings with a large aspect ratio. Never skip the one-way check just because two-way shear looked fine, especially on elongated footings.

#355 · Foundations and Retaining Structures

Where is the critical section for one-way (beam) shear in a footing?

Answer

One-way shear is checked at a distance d from the face of the column, taken across the full width of the footing. This treats the footing like a wide cantilevered beam loaded by upward soil pressure. It is a different location than the $d/2$ offset used for two-way shear.

#356 · Foundations and Retaining Structures

Where is the critical section for flexural design in a spread footing, and why?

Answer

The critical moment section is taken at the face of the column, not at d or $d/2$. Footings are treated as cantilevers bending upward from the column face under net factored soil pressure. Reinforcement is then designed with standard ACI flexural equations using that factored moment.

#357 · Foundations and Retaining Structures

A footing carries a total service load of 250 kips on soil with 4,000 psf allowable bearing pressure. What area is required and what size would you pick?

Answer

Required area equals 250,000 lb divided by 4,000 psf, which is 62.5 square feet. The square root of 62.5 is about 7.91 ft, so a designer rounds up to an 8 ft by 8 ft footing, giving 64 square feet. Checking, 250 kips over 64 sq ft equals 3,906 psf, which is under the 4,000 psf allowable.

#358 · Foundations and Retaining Structures

What is the main design goal for a combined footing supporting two columns?

Answer

The centroid of the footing area should line up with the resultant of the applied column loads. Matching centroid and resultant produces uniform bearing pressure across the footing. If they do not line up, the footing tips and one edge sees much higher pressure than the other.

#359 · Foundations and Retaining Structures

When would a rectangular combined footing be shaped asymmetrically versus a trapezoidal shape be used?

Answer

A rectangular combined footing is made asymmetric, projecting further on the lighter column's side, when the loads are fairly similar but an edge column has a property line restriction. A trapezoidal footing, wider at one end, is used when the two column loads differ significantly, since the extra width on the heavy end shifts the area centroid to match the load resultant.

#360 · Foundations and Retaining Structures

Why would a designer choose a mat (raft) foundation instead of individual spread footings?

Answer

Mats are used when soil bearing capacity is low, column loads are heavy, or individual footing areas would be so large they overlap each other. A mat spreads the whole building's load over one large slab covering the entire footprint. This lowers the average bearing pressure compared to many separate footings.

#361 · Foundations and Retaining Structures

What is the difference between the rigid method and the flexible method of mat foundation analysis?

Answer

The rigid method treats the mat as infinitely stiff, so bearing pressure varies linearly as P/A plus or minus Mc/I . The flexible method uses finite element analysis, letting the mat deflect and pressure redistribute realistically. The rigid, hand-calculation method is what this exam expects since a full finite element run does not fit a timed problem.

#362 · Foundations and Retaining Structures

True or false: a mat foundation is automatically safe in punching shear because it is so massive.

Answer

False. Punching shear checks around each column still apply on a mat exactly like on an individual footing, using the same $d/2$ critical section and ACI 318-14 two-way shear equations. Interior columns with a shear perimeter closed on all four sides are usually fine, but edge and corner columns with an open perimeter on one or two sides usually govern and need attention first.

#363 · Foundations and Retaining Structures

What two mechanisms transfer load from a deep foundation pile into the surrounding soil?

Answer

Piles transfer load through skin friction along the shaft, through end bearing at the pile tip, or through a combination of both. On this exam, the total geotechnical capacity of a pile is normally given directly as a design value in the problem. The structural engineer's job is designing the pile itself and the pile cap, not deriving capacity from soil borings.

#364 · Foundations and Retaining Structures

How is load distributed to individual piles in a pile group carrying an axial load P and a moment M ?

Answer

Load on any pile equals P divided by the number of piles, plus or minus M times that pile's distance from the group centroid, divided by the sum of distance squared for all piles in the group. This is the same elastic method used for eccentric bolt groups, just applied to a pile pattern instead of bolts. It is worth recognizing the formula is functionally identical between the two topics.

#365 · Foundations and Retaining Structures

How are pile caps designed structurally compared to ordinary spread footings?

Answer

Pile caps use nearly the same checks as spread footings: two-way shear at $d/2$ from the column face, one-way shear at d from the column face, and flexure at the column face. One extra wrinkle is that the two-way shear perimeter can be truncated when pile spacing is close, since nearby piles change how shear transfers across the perimeter.

#366 · Foundations and Retaining Structures

If a pile falls entirely inside the assumed two-way shear critical section of a pile cap, how is its reaction treated?

Answer

That pile's full reaction is typically excluded from causing shear at that critical section, because its load transfers more directly to the column without crossing the assumed perimeter. This is a detail the reference handbook will not flag automatically. Watch pile layout exhibits carefully for piles sitting inside the $d/2$ offset from the column face.

#367 · Foundations and Retaining Structures

What is a drilled shaft (caisson) and how is it structurally designed?

Answer

A drilled shaft is a large-diameter deep foundation drilled and cast in place, typically used for very heavy column loads or where pile-driving vibration cannot be tolerated near existing structures. It is designed under axial load, lateral load, and moment as a standard reinforced concrete column per ACI 318-14. The key difference from an ordinary column is that soil provides lateral support along its length instead of open air.

#368 · Foundations and Retaining Structures

What three stability checks must every cantilever retaining wall footing satisfy?

Answer

A cantilever retaining wall footing must be checked for overturning, sliding, and bearing pressure. All three checks use unfactored, service-level loads and forces because they are geotechnical stability checks, not concrete strength checks. Mixing in factored loads here is a common exam trap.

#369 · Foundations and Retaining Structures

How is the overturning factor of safety for a retaining wall calculated, and what value is typically required?

Answer

The overturning factor of safety equals the resisting moment (from the weight of the wall, footing, and soil above the heel, taken about the toe) divided by the overturning moment (from lateral earth pressure, taken about the same toe point). Standard practice generally requires this ratio to be at least 1.5. Both moments are calculated using service-level, unfactored forces.

#370 · Foundations and Retaining Structures

What forces make up the sliding stability check for a retaining wall, and what factor of safety is typical?

Answer

Resisting force is base friction, equal to the vertical load times a friction coefficient, plus any passive pressure resistance in front of the wall if the problem allows it to be counted. Driving force is the horizontal component of the lateral earth pressure. A minimum factor of safety of 1.5 against sliding is the typical target, matching the overturning requirement.

#371 · Foundations and Retaining Structures

What is the middle third rule for an eccentrically loaded retaining wall footing?

Answer

The middle third rule checks whether the load eccentricity stays within $B/6$ of the footing centerline, where B is the footing width. If eccentricity is within $B/6$, the bearing pressure distribution is trapezoidal over the full footing width. If eccentricity exceeds $B/6$, soil cannot resist the resulting negative pressure, so the distribution becomes triangular over a reduced effective width instead.

#372 · Foundations and Retaining Structures

A retaining wall footing is 8 ft wide, carries 12 kips per foot of wall at 1.2 ft eccentricity. Is the resultant within the middle third, and what are the max and min bearing pressures?

Answer

$B/6$ equals 8 divided by 6, which is 1.33 ft, and 1.2 ft is less than that, so the resultant stays within the middle third and the full trapezoidal distribution applies. Maximum pressure equals $(12/8)$ times $(1 + 6 \text{ times } 1.2 \text{ divided by } 8)$, which is 1.5 times 1.9, or 2.85 ksf. Minimum pressure equals 1.5 times $(1 \text{ minus } 0.9)$, or 0.15 ksf, and both values being positive confirms the middle third assumption was valid.

#373 · Foundations and Retaining Structures

How does a gravity retaining wall resist lateral soil pressure differently from a cantilever wall?

Answer

A gravity wall resists lateral pressure through sheer mass rather than bending stiffness in a thin stem. A cantilever wall transfers load through bending down a vertical stem into a footing with a soil wedge sitting on the heel. Both wall types still use the same overturning, sliding, and bearing checks, just with different resisting moment sources.

#374 · Foundations and Retaining Structures

What are the three lateral earth pressure states and what wall movement produces each one?

Answer

Active pressure develops when a wall moves slightly away from the retained soil, letting the soil expand and mobilize shear strength that reduces pressure on the wall. Passive pressure develops when a wall moves into the soil, compressing it and producing much higher pressure than active. At-rest pressure applies when the wall cannot move in either direction, such as a rigidly braced basement wall before backfill has a chance to move it.

#375 · Foundations and Retaining Structures

Which lateral pressure state is assumed for a normal free-standing cantilever or gravity retaining wall?

Answer

A normal free-standing retaining wall is assumed to develop active pressure, since it can deflect outward slightly under load. This is the lower of the three pressure states and is the standard default assumption unless the exhibit shows a rigid, unyielding wall condition. A braced basement wall, by contrast, is assigned at-rest pressure since it cannot move.

#376 · Foundations and Retaining Structures

What is the Rankine formula for the active earth pressure coefficient K_a , and what does it depend on?

Answer

K_a equals $\tan^2$ of $(45 \text{ minus } \phi/2)$, where ϕ is the soil's friction angle. The Rankine method depends only on friction angle and assumes a vertical wall back, a horizontal ground surface, and no wall friction. This formula and worked tables are available in the reference handbook, so it does not need to be memorized cold.

#377 · Foundations and Retaining Structures

What is the Rankine formula for the passive earth pressure coefficient K_p ?

Answer

K_p equals $\tan^2(45 + \phi/2)$, where ϕ is the soil's friction angle. K_p is always larger than K_a for the same friction angle, often by a factor of ten or more. Using K_p where K_a belongs, or vice versa, is a classic way to get a wildly wrong pressure result.

#378 · Foundations and Retaining Structures

How does the Coulomb method for lateral earth pressure differ from the Rankine method?

Answer

Coulomb's method accounts for wall friction, a sloped wall back, and a sloped backfill surface, producing different coefficients than Rankine for the same friction angle. Rankine assumes a vertical wall back, horizontal backfill, and no wall friction. Seeing a sloped backfill or battered wall face in an exhibit is the visual cue that Rankine's simplifying assumptions no longer strictly apply.

#379 · Foundations and Retaining Structures

What shape is the lateral earth pressure distribution behind a retaining wall with no water table or surcharge, and why?

Answer

The distribution is triangular, increasing linearly with depth, because pressure at depth z equals the earth pressure coefficient times soil unit weight times z . Since unit weight and coefficient are constant, pressure grows linearly as z increases. This is exactly analogous to a hydrostatic pressure triangle.

#380 · Foundations and Retaining Structures

What is the total lateral earth pressure resultant force per foot of wall for a triangular pressure distribution, and where does it act?

Answer

The resultant force equals one half times the earth pressure coefficient times the soil unit weight times the height squared, which is just the area of the pressure triangle. It acts at one third of the wall height up from the base, at the centroid of that triangular distribution. This resultant location matters directly for the overturning moment calculation.

#381 · Foundations and Retaining Structures

How does a surcharge load on the ground surface behind a retaining wall change the lateral pressure distribution?

Answer

A surcharge, such as a parking area or adjacent building load sitting on the backfill surface, adds a uniform rectangular pressure block equal to the earth pressure coefficient times the surcharge pressure. This rectangular block sits on top of the triangular soil pressure and stays constant with depth. It does not increase with depth the way the soil's own triangular pressure does.

#382 · Foundations and Retaining Structures

How does a water table behind a retaining wall change the lateral pressure calculation?

Answer

Below the water table, use the buoyant (effective) unit weight of soil for the earth pressure calculation instead of the total unit weight, since the water reduces the effective soil weight. Separately, add full hydrostatic water pressure using water's own unit weight, about 62.4 pcf, regardless of the soil's friction angle. Forgetting the separate hydrostatic term is one of the most common ways to get an otherwise correct retaining wall problem completely wrong.

#383 · Foundations and Retaining Structures

Why can a footing's column load include more than just axial force from the superstructure?

Answer

Column loads reaching a footing often already include moment transferred down from lateral load resisting elements like shear walls, braced frames, or moment frames that terminate at that foundation. The problem statement may state this base moment directly, or it may need to be derived by tracing the load path down from the lateral system. Reading the scenario carefully to see whether moment is already included is essential.

#384 · Foundations and Retaining Structures

What is uplift, and when does it commonly become a governing foundation design issue?

Answer

Uplift is a net upward force at a footing caused by overturning of a shear wall or braced frame under seismic or wind loading. A footing needs enough dead load, its own weight plus soil surcharge, to resist that net uplift, or it needs a tension connection like a pile or helical anchor sized for the force. This is especially common at the base of lateral load resisting elements under wind or seismic overturning.

#385 · Foundations and Retaining Structures

Which ASCE 7-16 load combination term is used to check whether dead load resists wind or seismic uplift, and why is it different from ordinary gravity design?

Answer

Load combinations that include a $0.6D$ term are used for the uplift check, since a reduced dead load conservatively tests whether it still counteracts overturning. Ordinary gravity design instead uses combinations like $1.2D$, which maximizes dead load. Using $1.2D$ by mistake when checking uplift would overstate the dead load's resisting effect and miss a real uplift failure.

#386 · Foundations and Retaining Structures

Why might differential settlement be a concern on a building with mixed foundation types?

Answer

Some columns might sit on spread footings while others sit on piles because of variable soil conditions across the site, and these foundation types settle by different amounts under load. That differential movement can induce extra forces into the superstructure that a purely gravity-load foundation design would never capture. Exhibits showing mixed foundation types or highly variable bearing values are a signal this issue is being tested.

#387 · Foundations and Retaining Structures

What is the single most common load-level trap in this domain, according to the chapter?

Answer

The most common trap is mixing load levels, such as using factored LRFD loads for a bearing pressure or overturning/sliding stability check, or using unfactored service loads for a concrete shear or flexure check. Bearing pressure and wall stability checks always use service loads, while concrete strength design always uses factored loads. Identify which load level a specific equation requires before pulling numbers from the scenario.

#388 · Foundations and Retaining Structures

Where does active pressure act on a retaining wall versus where passive pressure acts?

Answer

Active pressure develops on the side being retained, behind the wall, where the wall can move away from the soil. Passive pressure only develops in front of a wall or footing, where soil resists the structure moving into it. A single problem can present both situations on opposite sides of the same wall, so track which side each coefficient applies to.

#389 · Foundations and Retaining Structures

Name the three different critical section locations used for shear and flexure in footings, mats, and pile caps.

Answer

Two-way (punching) shear is checked at $d/2$ from the column face. One-way (beam) shear is checked at d from the column face. Flexure is checked at the face of the column itself. Mixing up these three locations can make a design look conservative in one check while actually being unconservative in another.

#390 · Foundations and Retaining Structures

Why is a water table or surcharge shown only in an exhibit's soil profile considered a common exam trap?

Answer

A water table symbol or surcharge callout may appear only as a label inside a plan or profile exhibit, not restated anywhere in the problem text. If it is missed, the lateral earth pressure calculation will use the wrong unit weight and skip the separate hydrostatic pressure term entirely. Always scan every exhibit's labels for these details before finalizing a lateral earth pressure calculation.

#391 · Foundations and Retaining Structures

Does the SE exam expect you to derive bearing capacity from Terzaghi's N_c , N_q , and N_{γ} factors?

Answer

No, that derivation is a geotechnical engineering task and is generally outside the structural depth's scope. Soil parameters like allowable bearing pressure, unit weight, and friction angle are almost always given directly in the problem scenario. The structural engineer's job is taking that given bearing pressure and sizing the footing, checking shear, checking flexure, and detailing reinforcement.

#392 · Foundations and Retaining Structures

What ACI 318-14 chapter governs footing design, and what three checks does it require beyond bearing?

Answer

ACI 318-14 Chapter 13 governs footing design. Beyond the initial bearing pressure sizing check, it requires two-way shear, one-way shear, and flexural design, each at its own distinct critical section. All three of these checks use factored loads and factored net soil pressure, unlike the bearing pressure check that set the footing's plan area.

#393 · Foundations and Retaining Structures

Is the SE Structural exam open book or closed book, and what reference material is provided?

Answer

The exam is closed book, but candidates are given an electronic NCEES reference handbook plus code chapters from ASCE 7-16, ACI 318-14, AISC 360, NDS 2018/SDPWS 2015, and TMS 402-16 loaded directly into the exam software. The exam is not testing whether an equation was memorized, but whether the right equation can be found fast and applied correctly. Time is better spent practicing navigation of the handbook than memorizing coefficient formulas.

#394 · Foundations and Retaining Structures

What three equations does ACI 318-14 provide for two-way shear capacity, and which one governs?

Answer

ACI 318-14 provides three two-way shear capacity equations, one based on β (the column aspect ratio), one based on α_s (related to column location, interior versus edge versus corner), and one simple expression equal to 4 times the square root of f'_c times b_o times d . The smallest of the three values normally governs the design. The reference handbook lists all three, so the key skill is knowing which situation triggers which equation, not memorizing them.

#395 · Foundations and Retaining Structures

What kind of footing shape typically results from a heavily eccentric combined footing where a trapezoid is used?

Answer

A trapezoidal combined footing is wider at the end nearer the heavier column and narrower at the end nearer the lighter column. This shape shifts the area centroid toward the heavy column's side to match the load resultant location. It is chosen over a simple rectangle when the two column loads differ significantly and site space is also limited on one side.

#396 · Foundations and Retaining Structures

For interior versus edge/corner columns on a mat foundation, which typically governs the two-way shear check and why?

Answer

Edge and corner columns typically govern because their shear perimeter is open on one or two sides instead of being closed on all four sides. Interior columns have a full closed perimeter around them and are usually fine. Always check edge and corner columns first on a mat foundation problem rather than assuming the interior columns control.

#397 · Foundations and Retaining Structures

What analogy connects pile group load distribution to a topic from the connections chapter?

Answer

Pile group load distribution under an axial load plus a moment uses the exact same elastic method as an eccentrically loaded bolt group. Load on any pile equals P over the number of piles, plus or minus M times its distance from the centroid divided by the sum of distance squared, which mirrors the bolt group formula exactly. Recognizing this shared structure means one method serves two different topics on the exam.

#398 · Foundations and Retaining Structures

Why does a battered (sloped) wall face or a sloped backfill surface signal a shift away from Rankine theory?

Answer

Rankine's method assumes a vertical wall back and a horizontal backfill surface with no wall friction considered. A sloped backfill or a battered wall face violates those simplifying assumptions, so Coulomb's more general method, which accounts for wall friction and sloped geometry, is the appropriate approach. Recognizing these visual cues in an exhibit is more important on this exam than hand-calculating Coulomb's equations from scratch.

#399 · Foundations and Retaining Structures

In the retaining wall bearing pressure formula P/A times $(1 \pm 6e/B)$, what does each term represent?

Answer

P/A is the average bearing pressure from the vertical load spread over the footing area. The term $6e/B$ captures the added pressure from eccentricity e relative to the footing width B , derived from the Mc/I flexural stress formula applied to a rectangular footing section. Adding the term gives maximum pressure at the toe, and subtracting it gives minimum pressure at the heel, valid only while the resultant stays within the middle third.

#400 · Foundations and Retaining Structures

Why is time better spent practicing handbook navigation than memorizing K_a , K_p , N_c , or N_q formulas for this exam?

Answer

The NCEES reference handbook already contains these formulas and worked coefficient tables, so recalling them from memory is not the tested skill. What is actually tested is recognizing which physical situation calls for which coefficient or equation, and applying it correctly under time pressure. Building that judgment, not memorizing arithmetic, is the stated purpose of this chapter.

#401 · Loads Lateral Earth Pressure and Analysis Methods

Which ASCE 7-16 chapter governs dead loads, and what does it give you?

Answer

Dead loads come from ASCE 7-16 Chapter 3. It gives material unit weights for concrete, steel, masonry, and wood, plus typical component weights like roofing and partitions. On the exam you look these values up in the handbook rather than recalling them from memory.

#402 · Loads Lateral Earth Pressure and Analysis Methods

What is the design live load for a typical office area per ASCE 7-16 Table 4.3-1?

Answer

Office areas are 50 psf. Corridors above the first floor in office buildings are 80 psf, and light storage is 125 psf. These specific numbers are worth memorizing because they appear constantly in scenario problems.

#403 · Loads Lateral Earth Pressure and Analysis Methods

What is a member's tributary area, in plain terms?

Answer

It is the floor or roof area that delivers load to that specific member. For an interior column on a regular grid, the tributary width in each direction is the average of the two adjacent spans. For an edge or corner column, only the spans where framing actually exists count.

#404 · Loads Lateral Earth Pressure and Analysis Methods

Write the ASCE 7-16 Section 4.7 live load reduction equation and define every variable.

Answer

L equals L_o times the quantity 0.25 plus 15 divided by the square root of KLL times AT . L_o is the unreduced live load from Table 4.3-1, KLL is the live load element factor, and AT is the member's tributary area in square feet.

#405 · Loads Lateral Earth Pressure and Analysis Methods

What KLL value applies to an interior column, and what does KLL represent physically?

Answer

An interior column uses KLL equal to 4. KLL is the live load element factor, and multiplying it by AT approximates the true influence area contributing load to that member, which is larger than the raw tributary area for a column.

#406 · Loads Lateral Earth Pressure and Analysis Methods

List the KLL values for an edge column without a cantilever slab, an interior beam, and a one-way slab.

Answer

Edge and corner columns without a cantilever slab condition use KLL equal to 4, the same as interior columns -- column position alone does not change this factor. Edge columns with a cantilever slab use KLL equal to 3, corner columns with a cantilever slab use KLL equal to 2, interior beams and edge beams without a cantilever slab use KLL equal to 2, and one-way slabs use KLL equal to 1. Only a cantilever-slab condition reduces a column's KLL below 4.

#407 · Loads Lateral Earth Pressure and Analysis Methods

What is the minimum KLL times AT threshold before live load reduction is allowed to apply at all?

Answer

The reduction only applies once KLL times AT is at least 400 square feet. Below that threshold you must design for the full unreduced live load L_o , no matter how the equation would otherwise come out.

#408 · Loads Lateral Earth Pressure and Analysis Methods

What is the floor on how far live load can be reduced for a member supporting only one floor versus two or more floors?

Answer

For a member supporting one floor, L cannot drop below 0.4 times L_o . For a member supporting two or more floors, L cannot drop below 0.5 times L_o . Forgetting to check this floor value after computing the equation is a very common exam mistake.

#409 · Loads Lateral Earth Pressure and Analysis Methods

Trap: can you apply the Section 4.7 live load reduction equation to roof live loads?

Answer

No. Roof live loads use an entirely separate reduction method under ASCE 7-16 Section 4.8, not the floor live load reduction equation from Section 4.7. Mixing the two methods is a classic exam trap.

#410 · Loads Lateral Earth Pressure and Analysis Methods

Are areas of public assembly or live loads over 100 psf eligible for the standard live load reduction?

Answer

Generally no. Areas of public assembly and live loads exceeding 100 psf are restricted from this type of reduction, since the assumption that full load rarely acts everywhere at once does not hold as well for those occupancies.

#411 · Loads Lateral Earth Pressure and Analysis Methods

An interior column has $AT = 720$ sq ft, $Lo = 50$ psf, $KLL = 4$. Find the reduced live load.

Answer

KLL times AT equals 2880, well above the 400 sq ft threshold, so reduction applies. The square root of 2880 is about 53.67, so 15 divided by that is 0.2795. Adding 0.25 gives 0.5295, times 50 psf gives L equal to about 26.5 psf, which is above the 0.4 times Lo floor of 20 psf, so 26.5 psf stands.

#412 · Loads Lateral Earth Pressure and Analysis Methods

Which ASCE 7-16 chapter governs snow loads, and what is the basic flat roof snow equation?

Answer

Chapter 7 governs snow loads. The flat roof snow load pf equals 0.7 times C_e times C_t times I_s times p_g , where p_g is the ground snow load, C_e is the exposure factor, C_t is the thermal factor, and I_s is the importance factor tied to Risk Category.

#413 · Loads Lateral Earth Pressure and Analysis Methods

What does the thermal factor C_t represent, and what value does it take for a normally heated structure?

Answer

C_t accounts for how well a structure retains heat, since a warmer roof melts snow faster. A heated structure uses C_t equal to 1.0, while an unheated structure gets a higher C_t because it retains more snow on the roof.

#414 · Loads Lateral Earth Pressure and Analysis Methods

What is the 0.7 constant in the flat roof snow equation, and why should you know it cold?

Answer

It is a fixed empirical reduction factor built into the p_f equation itself, applied on top of the four separate factor lookups. You should have the 0.7 and the equation's arrangement memorized so you are not pulling the whole formula from a code page while also juggling four table lookups under time pressure.

#415 · Loads Lateral Earth Pressure and Analysis Methods

What is the minimum snow load rule p_m in ASCE 7-16 Section 7.3.4, and when does it govern?

Answer

For low-slope roofs where p_g is 20 psf or less, the design flat roof snow load cannot be taken as less than p_g times I_s , called p_m . This minimum governs whenever it exceeds the standard p_f equation's result, which happens often at low p_g sites because the 0.7 factor pulls p_f below p_g times I_s .

#416 · Loads Lateral Earth Pressure and Analysis Methods

Trap: is p_m a floor applied to every snow load calculation?

Answer

No. p_m only overrides p_f in the specific low-slope, low p_g scenario described in Section 7.3.4, and even then it only governs when p_m turns out to be the larger of the two values. Treating p_m as a universal floor on all snow calculations is a common mistake.

#417 · Loads Lateral Earth Pressure and Analysis Methods

How is sloped roof snow load found once you have p_f ?

Answer

Sloped roof snow load multiplies p_f by a roof slope factor C_s , which depends on the roof slope, the thermal condition, and how slippery the roof surface is. C_s is looked up from a chart in the handbook.

#418 · Loads Lateral Earth Pressure and Analysis Methods

What triggers a snow drift check between a lower roof and an adjacent higher roof or rooftop unit?

Answer

Wind blows snow off the upper surface and piles it against the step down to the lower roof. You compute h_b , the balanced snow depth on the lower roof, and h_c , the actual clear height difference after that balanced snow is sitting there. If h_c divided by h_b is greater than 0.2, drift must be considered.

#419 · Loads Lateral Earth Pressure and Analysis Methods

How do you calculate the balanced snow depth h_b and the snow density γ used in drift calculations?

Answer

h_b equals p_f divided by γ . The snow density γ equals 0.13 times p_g plus 14, capped at a maximum of 30 pcf. This density is what converts the balanced snow load p_f into an actual physical depth.

#420 · Loads Lateral Earth Pressure and Analysis Methods

What happens to the drift calculation if the computed drift height h_d would exceed the clear height h_c ?

Answer

If h_d would exceed h_c , the drift is truncated at h_c , and instead of increasing the height further, the drift width increases using a different equation to preserve the same total drift area. If h_d is less than h_c , you simply use h_d as calculated, with no truncation needed.

#421 · Loads Lateral Earth Pressure and Analysis Methods

How is the drift surcharge load intensity distributed across the lower roof?

Answer

The surcharge intensity at the wall equals h_d times γ , added on top of the balanced load p_f . This peak value tapers linearly down to zero over a drift width w equal to 4 times h_d , unless h_d was truncated at h_c , in which case a different width formula applies.

#422 · Loads Lateral Earth Pressure and Analysis Methods

A roof step has $p_g = 30$ psf, $C_e = C_t = I_s = 1.0$, and an 8 ft step height. Find p_f , γ , h_b , and h_c .

Answer

p_f equals 0.7 times 30, or 21 psf. γ equals 0.13 times 30 plus 14, or 17.9 pcf. h_b equals 21 divided by 17.9, or about 1.17 ft. h_c equals the 8 ft step minus h_b , or about 6.83 ft, and since h_c over h_b is about 5.84, well above 0.2, drift must be considered.

#423 · Loads Lateral Earth Pressure and Analysis Methods

Which ASCE 7-16 chapter governs rain loads, and what scenario does the design rain load represent?

Answer

Chapter 8 governs rain loads. The design rain load represents the depth of water that would accumulate on a flat or low-slope roof if the primary drains were blocked and only the secondary overflow drainage system were working, at that system's flow rate, computed as a static hydraulic head.

#424 · Loads Lateral Earth Pressure and Analysis Methods

What is ponding instability, and why is it more dangerous than a simple rain load calculation?

Answer

Ponding instability is a feedback loop where rainwater collects in a shallow bowl as flexible roof framing deflects under the water's weight, and that added water causes more deflection, which collects more water, potentially progressing toward collapse. It is a stiffness criterion on the framing, not just a load magnitude to add into a combination.

#425 · Loads Lateral Earth Pressure and Analysis Methods

Under what conditions does a ponding instability check become critical?

Answer

It becomes critical whenever roof slope is very low, under about 1/4 inch per foot, and primary drainage could plausibly become blocked. The exam trap is treating rain load like an ordinary gravity load without recognizing the scenario is actually testing whether you flag ponding instability as a distinct concern.

#426 · Loads Lateral Earth Pressure and Analysis Methods

Which ASCE 7-16 chapter and sections give the LRFD and ASD basic load combinations?

Answer

Chapter 2 gives both sets. LRFD strength design combinations are in Section 2.3, and ASD allowable stress design combinations are in Section 2.4. Both sets are printed in full in the exam handbook, so you do not need to recall the coefficients from memory.

#427 · Loads Lateral Earth Pressure and Analysis Methods

List the basic LRFD load combinations from ASCE 7-16 Section 2.3.

Answer

They are $1.4D$; $1.2D + 1.6L + 0.5(L_r \text{ or } S \text{ or } R)$; $1.2D + 1.6(L_r \text{ or } S \text{ or } R) + (1.0L \text{ or } 0.5W)$; $1.2D + 1.0W + 1.0L + 0.5(L_r \text{ or } S \text{ or } R)$; $1.2D + 1.0E + 1.0L + 0.2S$; and $0.9D + 1.0W$ or $0.9D + 1.0E$.

#428 · Loads Lateral Earth Pressure and Analysis Methods

Why does wind get a 1.0 load factor in ASCE 7-10 and later LRFD combinations instead of the older 1.6 factor?

Answer

Since ASCE 7-10, wind pressures from Chapters 26 through 31 are already given at the strength or ultimate level, unlike older codes where the tabulated pressures were service-level and needed a 1.6 factor. Applying 1.6 anyway would double count a load factor already baked into the reference pressure.

#429 · Loads Lateral Earth Pressure and Analysis Methods

Which combinations typically govern net uplift or overturning checks, and why?

Answer

The 0.9D combinations, either $0.9D + 1.0W$ or $0.9D + 1.0E$, typically govern because they minimize the stabilizing dead load while keeping the full lateral or wind demand. This combination usually controls anchor bolt tension and connection uplift capacity, even though it never governs a beam's maximum bending moment.

#430 · Loads Lateral Earth Pressure and Analysis Methods

A roof beam has $D = 120$ plf, $L_r = 72$ plf, and strength-level wind uplift $W = -168$ plf. Find the governing gravity combination and the uplift check.

Answer

$1.2D + 1.6L_r$ gives 144 plus 115.2, or 259.2 plf, the largest downward gravity value. The uplift check $0.9D + 1.0W$ gives 108 minus 168, or negative 60 plf, meaning a net upward force of 60 plf that connections must resist, even though gravity governs the beam's bending design.

#431 · Loads Lateral Earth Pressure and Analysis Methods

Trap: does the same load combination govern every design check on a single roof member?

Answer

No. A very common exam pattern gives dead, roof live, and wind loads on one member and asks which combination produces the largest downward force versus which produces net uplift. These are two different combinations with two different answers, so both must be checked separately rather than assuming one governs everything.

#432 · Loads Lateral Earth Pressure and Analysis Methods

What is tributary area analysis good for, and what does it ignore?

Answer

It is the fastest and most common method for gravity load takedown on regular framing, assigning each member a simple geometric area of load. It ignores continuity effects between spans, so it is an approximation appropriate for preliminary design and most gravity columns, not an exact elastic solution.

#433 · Loads Lateral Earth Pressure and Analysis Methods

What four conditions must be met before you can use ACI 318-14 Section 6.5 approximate moment and shear coefficients?

Answer

There must be two or more spans, uniformly distributed load, spans differing by no more than 20 percent from each other, and unfactored live load not exceeding three times unfactored dead load. If any condition fails, a real elastic analysis is required instead.

#434 · Loads Lateral Earth Pressure and Analysis Methods

In the ACI approximate coefficient method, what is the general form of the moment equation?

Answer

Moment equals a coefficient times w_u times l_n squared, where l_n is the clear span and w_u is the factored uniform load. The coefficient depends on whether you want positive span moment or negative moment at an interior or discontinuous end support, and on whether that end is restrained or unrestrained.

#435 · Loads Lateral Earth Pressure and Analysis Methods

What ACI coefficient applies to positive moment in an interior span, and what is it for an end span with an unrestrained discontinuous end?

Answer

Positive moment in an interior span uses l_n squared over 16. An end span with an unrestrained discontinuous end, meaning it simply rests on a wall, uses the larger coefficient l_n squared over 11.

#436 · Loads Lateral Earth Pressure and Analysis Methods

How does the ACI positive moment coefficient for an end span change if the discontinuous end is built integrally with a support instead of resting on a wall?

Answer

If the discontinuous end is restrained, meaning it is built integrally with a column or spandrel rather than simply resting on a wall, the coefficient becomes l_n squared over 14 instead of l_n squared over 11 for the unrestrained case.

#437 · Loads Lateral Earth Pressure and Analysis Methods

What ACI coefficients apply to negative moment at the first interior support, and how does the span count change it?

Answer

Negative moment at the exterior face of the first interior support uses l_n squared over 9 when there are exactly two spans, but l_n squared over 10 when there are more than two spans. Negative moment at other interior support faces uses l_n squared over 11.

#438 · Loads Lateral Earth Pressure and Analysis Methods

A four-span continuous beam has $l_n = 20$ ft and $w_u = 3.5$ kips/ft. Find the positive interior span moment and the negative moment at the first interior support.

Answer

w_u times l_n squared equals 3.5 times 400, or 1400 kip-ft. Positive interior span moment equals 1400 divided by 16, or 87.5 kip-ft. Negative moment at the exterior face of the first interior support, since there are more than two spans, equals 1400 divided by 10, or 140 kip-ft.

#439 · Loads Lateral Earth Pressure and Analysis Methods

Trap: a continuous beam has 4 spans but one span is 30 percent longer than its neighbors. Can you still use the ACI 318-14 Section 6.5 coefficients?

Answer

No. One of the required conditions is that spans differ by no more than 20 percent from each other, and a 30 percent difference violates it. When a scenario deliberately breaks one of these conditions, the approximate coefficients are the wrong tool and a real elastic analysis is expected instead.

#440 · Loads Lateral Earth Pressure and Analysis Methods

What is moment distribution used for, and what does it model conceptually?

Answer

Moment distribution is the classical hand method for analyzing frames and continuous beams that do not qualify for the ACI approximate coefficients. It models how stiffness and distribution factors share moment between members meeting at a common joint.

#441 · Loads Lateral Earth Pressure and Analysis Methods

What does an influence line describe, and why does it matter for live load placement?

Answer

An influence line describes how a response quantity, such as a reaction or the moment at one point, changes as a unit load moves across a structure. It matters because moving live loads require checking multiple load positions to find the one that maximizes a particular effect, not just a single static load case.

#442 · Loads Lateral Earth Pressure and Analysis Methods

What earth pressure coefficient applies to a basement wall braced top and bottom that cannot move at all?

Answer

A wall that cannot move at all uses at-rest pressure with coefficient K_0 , commonly approximated as $1 - \sin \phi$ for normally consolidated soil. This is the correct case for a basement wall restrained by both a floor diaphragm above and a slab or footing below.

#443 · Loads Lateral Earth Pressure and Analysis Methods

Rank the three earth pressure coefficients K_0 , K_a , and K_p from smallest to largest.

Answer

Active pressure coefficient K_a is smallest, at-rest coefficient K_0 is in the middle, and passive pressure coefficient K_p is always the largest of the three. Active applies to a wall free to move away from the soil, and passive resists a wall being pushed into the soil.

#444 · Loads Lateral Earth Pressure and Analysis Methods

How is lateral earth pressure at a given depth calculated, and how do you handle a surcharge like an adjacent parking area?

Answer

Each pressure is triangular, increasing linearly with depth, computed as the appropriate K coefficient times the soil unit weight times depth. A surcharge, such as a parking area or an adjacent footing, is added as an additional uniform or triangular pressure component on top of that triangular distribution.

#445 · Loads Lateral Earth Pressure and Analysis Methods

Trap: why is it wrong to design a basement wall restrained at both top and bottom using active earth pressure?

Answer

Active pressure requires the wall to rotate or translate enough away from the soil to mobilize that lower pressure condition. A wall restrained at the top by a floor diaphragm and at the bottom by a slab or footing cannot deflect that much, so it must be designed for the meaningfully larger at-rest pressure instead.

#446 · Loads Lateral Earth Pressure and Analysis Methods

Where does a cantilever retaining wall stem fit into the Vertical Buildings Depth exam, given that lateral system design belongs to the Lateral component?

Answer

Even though deep lateral system design is tested on the separate Lateral exam component, foundation and retaining wall elements on the Buildings exam still require lateral earth pressure to be treated as a load input feeding a vertical member's design, such as a basement wall or a cantilever retaining stem.

#447 · Loads Lateral Earth Pressure and Analysis Methods

Why should you practice scanning ASCE 7-16 handbook tables quickly rather than memorizing every unit weight or factor value?

Answer

On the real exam you have the ASCE 7-16 code chapters and the NCEES reference handbook available in the exam software, so equations and most factor tables do not need to be recalled from memory. The real skill is speed: knowing which table or section to open, which variable goes where, and sanity checking the resulting number.

#448 · Loads Lateral Earth Pressure and Analysis Methods

What judgment calls does the handbook NOT make for you, according to this chapter's closing summary?

Answer

The handbook gives you equations, factor tables, and code text, but it never tells you which load combination controls, which KLL value applies to your specific member type, or whether a wall is restrained or free to translate. Those judgment calls are yours to make, and they are exactly what the exam is measuring.

#449 · Loads Lateral Earth Pressure and Analysis Methods

Why is L_o different from L in the live load reduction equation, and where does each value come from?

Answer

L_o is the unreduced base live load pulled straight from ASCE 7-16 Table 4.3-1 for the occupancy in question. L is the reduced design live load produced after applying the Section 4.7 equation, and it is L , not L_o , that you actually use in the load combinations once reduction is permitted.

#450 · Loads Lateral Earth Pressure and Analysis Methods

A scenario gives you a low-slope roof with $p_g = 15$ psf and a computed $p_f = 9$ psf with $I_s = 1.0$. Which value governs, and why?

Answer

Since p_g is 20 psf or less, the minimum p_m applies, equal to p_g times I_s , or 15 psf. Because p_m of 15 psf exceeds the computed p_f of 9 psf, p_m governs as the design flat roof snow load, illustrating exactly why the 0.7 factor can pull p_f below the minimum at low p_g sites.

#451 · Connections and Joints

What is the fundamental difference between a bearing-type bolted connection and a slip-critical bolted connection under AISC 360 Chapter J?

Answer

A bearing-type connection allows the bolts to slip into bearing against the hole wall, so the governing limit states are bolt shear and material bearing or tearout. A slip-critical connection cannot tolerate slip at all, so it is checked against friction resistance from bolt pretension using a slip coefficient and hole factor per Section J3.8. Slip-critical connections are used where slip would cause fatigue, cladding damage, or oversized hole movement problems.

#452 · Connections and Joints

For an eccentrically loaded bolt group, name the two solution methods and state which is faster on exam day.

Answer

You can use the elastic vector method or the instantaneous center of rotation (IC) method with the tables in AISC Manual Part 7. The IC method is faster on exam day because the coefficient C is already tabulated for common geometries. The elastic method is more conceptually transparent but slower by hand, so treat it as a backup or sanity check rather than a first choice.

#453 · Connections and Joints

A 4 bolt bracket connection has bolts in a single vertical row at 3 inch spacing, loaded by 40 kips applied 5 inches eccentric. Using the elastic method, what is the resultant force on the critical bolt?

Answer

The sum of squared distances from the centroid is 45 in squared, giving a moment force of 200 kip-in times 4.5 divided by 45, or 20 kips horizontal. Direct shear is 40 divided by 4, or 10 kips vertical. Combining by square root of sum of squares gives the square root of 500, about 22.4 kips on the critical bolt. That value then gets checked against the single bolt shear capacity in the Manual tables.

#454 · Connections and Joints

Why can a bolted connection fail even when the bolts themselves have plenty of remaining shear capacity?

Answer

The connected material can govern through bearing or tearout at the bolt holes, per AISC 360 Section J3.10. A tight edge distance or clear spacing reduces the tearout equation dramatically because that equation depends directly on the clear distance to the edge or next hole times thickness and F_u . A scenario giving an edge distance close to code minimum is a strong hint that tearout governs.

#455 · Connections and Joints

What is block shear and where does it commonly appear on the exam?

Answer

Block shear is a rupture limit state in the connected material where a block of plate tears out along a combined shear and tension path, governed by AISC 360 Section J4.3. It shows up constantly in coped beam connections and bracket plates. The equation takes the larger of the shear yielding or shear rupture term added to the tension rupture term, and getting the failure block geometry wrong throws off every area in the calculation at once.

#456 · Connections and Joints

Why does a CJP groove weld usually not need its capacity computed on the exam?

Answer

A complete joint penetration groove weld is typically designed to develop the full strength of the connected base material. So an exam question involving a CJP weld is usually testing whether you recognize that the weld itself does not govern the connection, rather than asking you to calculate its strength.

#457 · Connections and Joints

State the fillet weld effective throat and nominal strength per unit length equations.

Answer

The effective throat of a fillet weld equals 0.707 times the leg size. Nominal strength per unit length equals 0.6 times the electrode strength F_{exx} times that effective throat. These two equations are the core tool for any fillet weld capacity check on the exam.

#458 · Connections and Joints

What common factor do candidates forget when checking fillet weld strength, and why does it matter?

Answer

AISC allows a 1.5 increase factor for fillet welds loaded transversely compared to longitudinally loaded welds. Forgetting this factor changes the computed strength depending on the load direction relative to the weld axis. Always identify which orientation the problem describes before pulling a strength value from the table.

#459 · Connections and Joints

An 8 inch long, single sided, 5/16 inch fillet weld is loaded concentrically in shear by 30 kips using E70 electrode. Find the LRFD design strength and confirm it is adequate.

Answer

Effective throat is 0.707 times 0.3125 inches, or 0.221 inches. Nominal strength per inch is 0.6 times 70 ksi times 0.221 inches, or 9.28 kips per inch. Total nominal capacity is 9.28 times 8 inches, about 74.3 kips. Applying phi of 0.75 gives a design strength of about 55.7 kips, which comfortably exceeds the 30 kip demand.

#460 · Connections and Joints

How does a partial joint penetration (PJP) groove weld differ from a CJP weld in design approach?

Answer

A PJP weld's effective throat is not the full plate thickness, it depends on the groove angle and welding process from a handbook table. Unlike a CJP weld, a PJP weld does not automatically develop full base metal strength. You must compute its capacity the same way as a fillet weld, using the effective throat times 0.6 F_{exx} .

#461 · Connections and Joints

Weld group eccentricity uses which reference tool for fast exam solving?

Answer

The IC method tables in AISC Manual Part 8 give a coefficient C for common weld group geometries such as L-shaped, C-shaped, and return welds, as a function of the eccentricity ratio. Practicing how to read the geometry ratios k and $a-l$ that index into these tables ahead of time is much faster than working the elastic vector method during the exam.

#462 · Connections and Joints

List the seven separate limit states that ACI 318-14 Chapter 17 requires you to check for a concrete anchor.

Answer

The limit states are steel strength in tension, concrete breakout in tension, pullout strength, concrete side-face blowout, steel strength in shear, concrete breakout in shear, and concrete pryout strength in shear.

#463 · Connections and Joints

What is the most common trap in concrete breakout capacity calculations using the CCD method?

Answer

Forgetting the cracked concrete assumption. Unless a problem explicitly states the concrete is uncracked at service loads, you must use the cracked concrete condition factors, which give noticeably lower breakout capacity than the uncracked case. The CCD method itself is built around a projected area ratio AN_c over AN_{co} times several modification factors for edge distance, eccentricity, cracking, and splitting reinforcement.

#464 · Connections and Joints

When must an anchor group be checked as one combined breakout surface instead of individual anchors added together?

Answer

When anchors are spaced closer than roughly three times the effective embedment depth ($3h_{ef}$), their individual breakout cones overlap and the group must be checked as one combined surface. This combined check almost always governs over simply summing individual anchor capacities, so always sketch the projected failure surface before doing the arithmetic.

#465 · Connections and Joints

A single cast-in headed anchor has hef of 6 inches. What is A_{Nco} , and is a second identical anchor 8 inches away independent or part of a group?

Answer

A_{Nco} for a single anchor equals 9 times hef squared, which is 9 times 36, or 324 square inches. The critical spacing $3h_{ef}$ is 18 inches, and 8 inches is less than that, so the two anchors share overlapping breakout cones and must be checked together as a group rather than as two independent anchors summed.

#466 · Connections and Joints

What is the key design difference between cast-in anchors and post-installed anchors on the SE exam?

Answer

Cast-in anchors such as headed studs or hooked bolts use pullout strength equations based on bearing at the head or hook resistance from ACI 318, not bond equations.

#467 · Connections and Joints

What limit state exists for adhesive anchors that does not exist for cast-in anchors, and what is it sensitive to?

Answer

Adhesive anchors have a bond strength limit state that cast-in anchors do not have. That bond strength is sensitive to installation temperature and hole cleaning. On the exam this usually appears as a bond stress value given directly in the problem, paired with an edge distance and spacing check using the same CCD breakout framework used for any other anchor.

#468 · Connections and Joints

True or false: a post-installed anchor problem always needs a completely different method than a cast-in anchor problem.

Answer

False. The breakout and steel strength equations are shared between post-installed and cast-in anchors. Only the bond or pullout term changes for adhesive or mechanical post-installed types, so do not assume the whole method changes.

#469 · Connections and Joints

In a wood connection joining a wood member to steel or concrete, what usually governs capacity, and why does this surprise candidates?

Answer

Wood connection capacity is almost always controlled by the wood member yielding around the fastener, not by the fastener steel failing. This surprises many candidates coming from a steel-heavy background who expect the metal fastener to be the weak link. NDS 2018 yield limit equations model failure modes like wood crushing and fastener bending with one or two plastic hinges, and the code automatically selects the governing mode once you input the right dowel bearing strengths.

#470 · Connections and Joints

Give an example of where wood connection design shows up in a cross-material context on this exam.

Answer

Common examples are a wood ledger bolted to a concrete wall or a wood post base plate anchored with post-installed anchors. NDS 2018 reference design values for bolts and lag screws already include adjustment factors for species, moisture, and load duration, summarized in the handbook's yield limit equations and adjustment factor tables.

#471 · Connections and Joints

What is the core capacity design philosophy tested across every material system in this chapter?

Answer

Connections should be designed so a ductile limit state controls while brittle limit states are pushed to be stronger than the ductile one. In steel this means plate yielding, a ductile mode, should govern before bolt or weld rupture, a brittle mode. In concrete anchorage it means steel yielding of the anchor should generally occur before a sudden concrete breakout failure.

#472 · Connections and Joints

Where does capacity design philosophy for anchors specifically appear in ACI 318-14?

Answer

Chapter 17 includes ductility requirements for anchors resisting seismic forces, favoring steel yielding of the anchor over a sudden brittle concrete breakout failure. This mirrors the same ductile-before-brittle philosophy applied to steel bolted and welded connections elsewhere in the code.

#473 · Connections and Joints

What does it mean when an exam problem gives you an overstrength factor or asks for a probable maximum capacity?

Answer

That phrasing is the cue that a capacity design check is being tested, as used in AISC 341 and the Seismic Design Manual for seismic force resisting system connections. The connection must be designed for forces amplified by the expected yield strength of the connected member, not just the code level demand, rather than a simple demand versus capacity comparison.

#474 · Connections and Joints

Name the most common numeric trap across steel, concrete, and wood connection problems on this exam.

Answer

Mixing up nominal strength, LRFD design strength (with the resistance factor applied), and ASD allowable strength (divided by the safety factor). These three numbers for the same connection can differ by 30 percent or more, so always confirm which one the question is actually asking for before finishing a calculation.

#475 · Connections and Joints

What should you do in the first thirty seconds of any connection problem on exam day, and why?

Answer

Identify every limit state in play, such as bolt shear, bearing, block shear, weld rupture, breakout, pullout, and wood bearing, and roughly rank which one is likely to control. This lets you calculate the governing case first instead of working every limit state in the order the problem happens to list them, which saves time under exam pressure. Also watch for one small changed word, like transverse versus longitudinal or cracked versus uncracked, that shifts which table or factor applies, and keep units written next to every number since ACI anchor equations use pounds and psi while other calculations use kips and ksi.

#476 · Conceptual Design Project Planning and Documentation

Why does the Conceptual Design and Project Planning domain reward judgment more than memorized formulas?

Answer

During the exam you have a searchable electronic reference handbook with the full codes, so any equation, table value, or deflection limit can be looked up in seconds. What you cannot look up is judgment, such as which structural system fits a given constraint or what your responsibility is when a field condition falls short. Study time is best spent on reasoning skills, not on memorizing values that are already on screen.

#477 · Conceptual Design Project Planning and Documentation

A parking garage owner wants long clear spans and minimal columns for car maneuvering. What structural systems fit this program?

Answer

Precast double tees or post tensioned concrete are the typical choices because both can achieve long clear spans efficiently while keeping columns out of drive aisles. The system choice should follow the owner's functional program first, not the engineer's personal preference. Span capability and open sightlines drive this decision more than raw material cost.

#478 · Conceptual Design Project Planning and Documentation

What is the main tradeoff of steel framing compared to cast in place concrete?

Answer

Steel is fast to erect, predictable in weight, and efficient for long spans, but it typically needs fireproofing and is more sensitive to field tolerance issues at connections. Cast in place concrete gives mass, fire resistance, and vibration damping without added fireproofing, but it is slower to build and more weather dependent during construction. Neither system is universally better, the right choice depends on schedule, fire rating, and site conditions.

#479 · Conceptual Design Project Planning and Documentation

Precast concrete moves fabrication off site and speeds erection. What risk does this concentrate?

Answer

Precast concentrates risk at connections and bearing details, which is exactly where many real coordination problems and RFIs show up. The speed benefit at the field comes with tighter tolerance demands on bearing seats, corbels, and connection hardware. An exam scenario about a bearing shortfall on a precast member is testing this exact tradeoff.

#480 · Conceptual Design Project Planning and Documentation

What codes govern wood lateral systems and masonry loadbearing wall systems on this exam?

Answer

Wood lateral systems are governed by SDPWS 2015 working alongside NDS 2018 for member design. Masonry loadbearing and lateral wall systems are governed by TMS 402/602-16. Masonry loadbearing walls work especially well for low to mid rise buildings with repetitive layouts like hotels or dormitories, where the walls double as the lateral system.

#481 · Conceptual Design Project Planning and Documentation

True or false: the cheapest structural system is always the one with the lowest material cost per pound.

Answer

False. A system that saves steel tonnage but requires six extra weeks of curing time can cost the owner more in financing and lost occupancy than a heavier but faster system. Cost decisions must weigh span, loading, schedule, site access, trade availability, and fire rating together. The exam expects a total project view, not a materials only comparison.

#482 · Conceptual Design Project Planning and Documentation

Why is placing a braced frame bay at a storefront entrance a coordination failure?

Answer

Braced frames, moment frames, and shear walls all take up space and interrupt column grids or door openings. A braced bay located where the architect needs an open storefront entrance cannot coexist with that program requirement. Lateral system layout has to be coordinated with architectural intent from the start, not overlaid afterward.

#483 · Conceptual Design Project Planning and Documentation

What creates the need for an expensive transfer structure at a podium level?

Answer

When shear walls cannot stack directly from roof to foundation because parking below has a different column and wall grid than the residential units above, the lateral load must transfer laterally at the podium level. This transfer requires a heavily loaded transfer diaphragm or transfer girder. Stacked, aligned shear walls are far more efficient and should be the design goal whenever the program allows it.

#484 · Conceptual Design Project Planning and Documentation

How does a flexible diaphragm distribute lateral load differently than a rigid diaphragm?

Answer

A flexible diaphragm, typical of wood panel sheathing or untopped metal deck, distributes lateral force to vertical elements based on tributary area. A rigid diaphragm, typical of concrete slabs, distributes load based on the relative stiffness of the vertical elements. Assuming rigid behavior over a long narrow flexible wood diaphragm is a classic conceptual error the exam likes to test.

#485 · Conceptual Design Project Planning and Documentation

What should you check first when a scenario shows an irregular building plan with a re-entrant corner?

Answer

Check for load path completeness before checking member sizes. Every lateral force needs a complete, continuous path from where it originates, typically a wall or diaphragm, down to the foundation, with no gaps at transitions such as a podium slab or a re-entrant corner. A member can be adequately sized and still fail the building if the load path around it is broken.

#486 · Conceptual Design Project Planning and Documentation

Most real structural coordination failures trace back to what root cause?

Answer

They usually trace back to depth conflicts between structure, ductwork, and finished ceilings, not to undersized members. The available vertical space between the finished floor below and finished ceiling above must fit the framing depth, the largest duct or pipe run, insulation, and the ceiling assembly all stacked together. When floor to floor height is fixed, framing depth becomes a design variable the engineer must actively manage.

#487 · Conceptual Design Project Planning and Documentation

Name two shallow framing strategies used to solve floor to floor depth conflicts without giving up span.

Answer

Post tensioned flat slabs and shallow long span trusses are two options, along with beams with web openings sized per AISC Design Guide 2. These strategies free up space for mechanical ducts and piping within a fixed floor to floor height. The engineer chooses these specifically when zoning or a repeating floor count within a height limit constrains available depth.

#488 · Conceptual Design Project Planning and Documentation

Why can a beam that is adequate in strength still cause a coordination problem for MEP and finishes?

Answer

A beam that deflects more than an attached partition, glazing system, or piping run can tolerate will crack finishes or overstress brittle connections even though the beam itself is perfectly adequate in strength. Deflection compatibility with attached nonstructural elements is a separate check from strength design. IBC Table 1604.3 gives standard limits, but recognizing which limit applies to a given finish condition is the judgment call being tested.

#489 · Conceptual Design Project Planning and Documentation

What deflection limits does IBC Table 1604.3 commonly give for typical floor members, and what is different for brittle finishes?

Answer

Common values are $L/360$ for live load and $L/240$ for total load on typical floor members. Members supporting brittle finishes need tighter limits than these standard values because brittle materials crack at smaller deflections. The table values are lookup material during the exam, the judgment is matching the right limit to the right finish condition.

#490 · Conceptual Design Project Planning and Documentation

Why is beam camber typically specified at about 80 percent of calculated dead load deflection rather than 100 percent?

Answer

Over cambering a beam that ends up carrying less dead load than calculated leaves a visible hump in the floor, which is much harder to fix than a small residual sag from under cambering. Specifying roughly 80 percent of the calculated dead load deflection, per common guidance in AISC Design Guide 3, builds in a margin against that overshoot. Getting the direction and magnitude conceptually right matters more on this exam than deriving the deflection equation.

#491 · Conceptual Design Project Planning and Documentation

What does a constructability review check beyond strength and serviceability?

Answer

It checks whether a design can actually be built in a sane sequence with normal equipment and normal tolerances. A connection requiring an overhead field weld in a tight shaft, or a transfer girder needing full shoring before three floors above it can be poured, can be technically correct on paper but still a scheduling disaster in the field. Constructability review catches these practical build sequence conflicts before they become RFIs.

#492 · Conceptual Design Project Planning and Documentation

When is a transfer beam supporting columns above often more heavily loaded than in its final condition?

Answer

During construction, while temporary shoring is still carrying load from the floors above, the transfer beam can see higher load than after the structure is complete and loads redistribute through the finished frame. Ignoring construction stage loading and checking only the final in-service condition is a common trap. Shoring and sequencing loads can govern a member even when the finished structure has plenty of capacity.

#493 · Conceptual Design Project Planning and Documentation

Who typically prepares erection engineering and temporary bracing plans for precast or steel, and who reviews them?

Answer

The contractor's specialty engineer usually prepares the erection engineering and temporary bracing plans as a separate deliverable from the final structural design. The engineer of record reviews these plans for compatibility with the permanent design rather than performing the erection engineering itself. Individual precast panels or steel members can be unstable before enough of the permanent lateral system is connected, which is why this separate deliverable exists.

#494 · Conceptual Design Project Planning and Documentation

What is the purpose of general notes on a structural drawing set?

Answer

General notes carry information that would be repetitive or impossible to show graphically on every sheet, such as design loads used, applicable code editions, material strengths, minimum bearing lengths, tolerances, and special inspection requirements. Reading general notes correctly and recognizing when a note conflicts with a detail is a real exam skill. A drawing set is incomplete if a detail contradicts what the general notes state.

#495 · Conceptual Design Project Planning and Documentation

Why does mixing elevation references, such as one detail dimensioned from finished floor and another from structural slab, cause problems?

Answer

Top of steel, top of slab, and top of footing elevations are typically referenced to a single project datum for consistency. Mixing references between details creates exactly the kind of ambiguity that generates RFIs during construction. A drawing set that is internally inconsistent in its elevation datum forces the field to guess, which is a documentation failure the exam expects you to catch.

#496 · Conceptual Design Project Planning and Documentation

What is the conceptual reason behind minimum bearing length and edge distance requirements for precast and masonry elements?

Answer

These minimums protect against local crushing and spalling under eccentric bearing. The PCI Design Handbook and TMS 402 give the specific lookup values, but knowing why those minimums exist is the conceptual knowledge the exam is really testing. A bearing question is never just about citing the code table, it is about understanding the failure mode the table is preventing.

#497 · Conceptual Design Project Planning and Documentation

Is shop drawing review by the engineer of record a re-design of the connection?

Answer

No. Shop drawing review is a check that the fabricator's detail matches the loads, geometry, and performance the EOR intended, not a re-design of the connection. The EOR is confirming conformance with design intent, not producing a new design for the fabricator. Confusing these two roles is a common misunderstanding of construction administration duties.

#498 · Conceptual Design Project Planning and Documentation

A contractor submits an RFI asking to accept a precast bearing length shorter than detailed. What must the EOR do?

Answer

The EOR must evaluate the actual as-built condition against the structural requirement, computing the real gap between required and provided, rather than simply approving the change to keep the schedule moving or rejecting it outright without analysis. Accepting a deficient condition without analysis exposes both the EOR and the public to real risk. Rejecting it without analysis when a defensible calculation could show adequate capacity wastes schedule and money unnecessarily.

#499 · Conceptual Design Project Planning and Documentation

If a field tolerance error is clearly the contractor's fault, does that shift responsibility for the as-built condition away from the engineer of record?

Answer

No. Even when the root cause is clearly a trade's mistake, the EOR is still responsible for determining whether the resulting as-built condition is structurally acceptable and for documenting that determination in writing. Means and methods belong to the contractor, but judging whether the finished condition is acceptable is always the EOR's call. This determination is typically documented through a formal RFI response or a supplemental sketch.

#500 · Conceptual Design Project Planning and Documentation

Forty precast members are erected with a deficient bearing condition before anyone notices, and the detail is then revised for remaining units. Is the project done?

Answer

No. Revising the detail going forward is necessary but not sufficient, since a fix applied to future units does not automatically resolve units already built. The forty already erected members still need their own evaluation and their own documented resolution. A scenario that only asks about fixing the connection detail in general is testing whether you catch this distinction about units already in place.